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Lithium Recovery Through Biohydrometallurgical Bioleaching

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Executive Summary

This plant is designed to recycle 15,000 tonnes of spent lithium-ion batteries each year and transform waste into valuable raw materials, including high-purity lithium carbonate, cobalt, nickel, and copper, while minimizing environmental impact and energy use. Using a cutting-edge biohydrometallurgical process, the facility combines advanced reactor control with robust safety measures to deliver efficient, sustainable recovery. Mass balances, performed with detailed process flow calculations and real-time monitoring, track every input, transformation, and output across each stage. To ensure precise management of resources and maximize recovery yields. Energy balances leverage electrical input data and passive system integration to optimize consumption and reduce the plant's carbon footprint, making the process not only effective but also cool, scalable, and eco-friendly for next-generation battery recycling.

Summary of Individual Contributions

Name of Student	List of Contributions	%*
Hekmat Kawas	- Calculations for mass balance - Calculations for energy balance - Research on the overall process - Process selection	25%
Regan Huynh	- Technological Survey - Equipment Selection - Hazardous Materials and Properties - Aerial View - Reaction Kinetics and Empirical Rules	25%
Shivam Patel	- Plant Safety - Logistics - Plant Location - Reaction Kinetics and Empirical Rules	25%
Maxim Plotnik	- Research on overall process calculations for mass balance - Introduction - Market Analysis - Conclusion	25%

*: Percentage of total contributions made by all group members

The undersigned attest that the aforementioned contributions are accurate.

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Nomenclature

Acronym	
Al	Aluminum
CAPEX	Capital Expenditure
Co	Cobalt
CSTR	Continuous Stirred-Tank Reactor
Cu	Copper
DEHPA	Di(2-ethylhexyl)phosphoric acid
EoL	End-of-life
Fe	Iron
HF	Hydrogen Fluoride
LIB	Lithium Ion Battery
Mn	Manganese
Ni	Nickel
OPEX	Operating Expenditure
ORP	Oxidation–Reduction Potential
PLS	Pregnant Leach Solution

Chapter 1. Introduction

The global shift toward electrification and decarbonization has accelerated demand for high-density energy storage, with lithium-ion batteries (LIBs) powering electric vehicles, consumer electronics, and renewable energy systems. As these technologies proliferate, so too does the volume of end-of-life (EoL) LIBs. This occurs due to hazardous waste streams containing flammable electrolytes, volatile organic compounds, and critical metals such as lithium, cobalt, and nickel. Landfilling is not a viable option due to risks of groundwater contamination and fire (Kim & Park, 2025), while discarding spent batteries represents a significant loss of strategic materials (Islam, 2022).

This report presents the design of a chemical processing plant capable of recycling 15,000 tonnes per year of spent LIBs. The facility is engineered to recover cobalt, nickel, lithium, copper, aluminum, and manganese as high-purity products suitable for reuse in the manufacturing of batteries. Key design challenges include managing feedstock heterogeneity, ensuring safe mechanical pretreatment, achieving selective metal separation, and minimizing environmental impact

A comprehensive technology survey was conducted to evaluate pyrometallurgical, hydrometallurgical, and biohydrometallurgical pathways. Based on performance metrics like recovery efficiency, energy use, reagent cost, safety, and sustainability, bioleaching was selected as the optimal process. The design integrates detailed process flow diagrams, mass and energy balances, safety analyses, and site selection criteria to deliver a scalable, economically viable, and environmentally responsible solution for LIB recycling (Roy et al., 2021).

1.1 The Selected Process

The technology selected for this plant design is a novel and highly efficient biohydrometallurgical process based on the work of Garcia et al. (2025). This approach was chosen over traditional pyrometallurgy and chemical hydrometallurgy due to its significantly reduced environmental impact, lower energy consumption, and exceptionally rapid kinetics. The process harnesses the metabolic activity of acidophilic bacteria (*Acidithiobacillus Ferrooxidans*) to generate a bio-oxidized ferric iron (Fe^{3+}) lixiviant, which, in synergy with the metallic components of the battery waste, achieves high metal recovery in minutes rather than days.

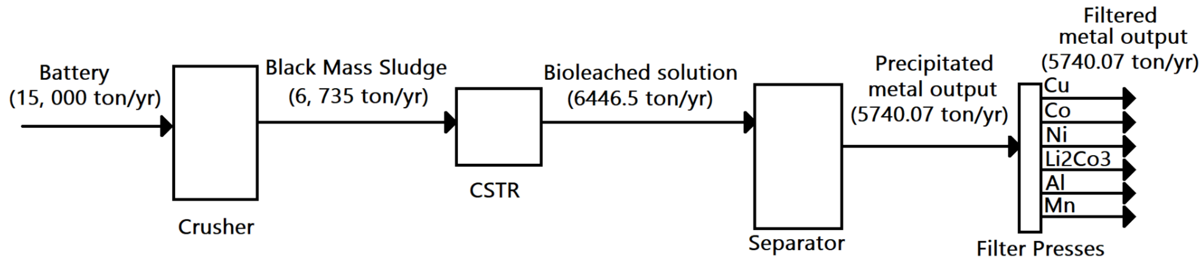

Figure 1.1: Block diagram of the process

As shown in the block diagram in Figure 1, the plant is designed to process 15,000 tonnes/year of spent lithium-ion batteries. An initial mechanical crushing and separation stage liberates the active materials, generating 6,735 tonnes/year of "Black Mass Sludge." This sludge is fed directly to the bioleaching CSTRs, where the valuable metals are dissolved. The subsequent separation and precipitation train recovers 5,740 tonnes/year of filtered metal products, including copper, cobalt, nickel, aluminum, manganese, and lithium (as lithium carbonate). The process operates at ambient temperature, with the primary energy input being electrical for pumps, crushing, and bioreactor aeration, rather than thermal energy for high-temperature smelting.

Chapter 2. Technological Survey

The rapid expansion of electric vehicles, portable electronics, and renewable energy storage has created an urgent need for sustainable recycling plants for lithium-ion batteries (LIBs). Designing such plants requires balancing technical efficiency, economic feasibility, and environmental sustainability, while also addressing safety risks and regulatory requirements. Current research and pilot-scale demonstrations reveal several technological pathways that can be used to recover valuable metals and close the loop on battery materials.

Historically, pyrometallurgy has been the dominant industrial method. This approach involves smelting batteries at very high temperatures, typically above 1000 °C, to recover cobalt, nickel, and copper as alloys. While robust and tolerant to mixed chemistries, pyrometallurgy is energy-intensive, generates toxic off-gases, and often results in the loss of lithium and aluminum in the slag. For plant designers, this pathway requires advanced gas-cleaning systems and slag treatment facilities, which add to both capital and operating costs (Roy et al., 2021; Cerrillo-Gonzalez et al., 2024).

Hydrometallurgy has emerged as a more selective and environmentally favourable alternative. In this method, pretreated black mass is subjected to acid leaching, using reagents such as sulfuric acid, hydrochloric acid, phosphoric acid, or increasingly, organic acids. The dissolved metals are then separated and purified through precipitation, solvent extraction, ion exchange, or electrodialysis. Hydrometallurgy offers high recovery rates and high-purity products with lower energy requirements compared to pyrometallurgy; however, it consumes large volumes of chemicals and water, generating wastewater that requires treatment. Recent innovations have improved its performance: Yang et al. (2017) demonstrated that mechanochemical activation of $LiFePO_4$ cathodes significantly enhances leachability, while Yang et al. (2018) showed that lithium can be selectively leached using acetic acid with hydrogen peroxide, achieving high recovery with minimal iron dissolution. Other studies have explored the use of biodegradable organic acids such as gluconic acid, which can achieve recovery rates above 95% for lithium, cobalt, and manganese (Lerchbammer et al., 2023). Electrodialysis has also been proposed as a way to reduce chemical consumption and improve selectivity in circular hydrometallurgical systems (Cerrillo-Gonzalez et al., 2024).

Biohydrometallurgy, also known as bioleaching, represents another promising pathway. This approach utilizes microorganisms such as Acidithiobacillus Ferrooxidans ferric ions, which dissolve metals from the black mass. Bioleaching is an environmentally benign process that requires little energy and can be scaled up using bioreactors. However, its kinetics are slow, and it is sensitive to pulp density and toxic electrolytes. Recent advances have demonstrated its potential for industrial applications. Garcia et al. (2025) reported that minimal pretreatment, combined with bio-oxidized Fe(III), enabled the recovery of 95% of copper, nickel, and aluminum within 20 minutes, and over 90% recovery of cobalt, manganese, and lithium within 30 minutes. Roy et al. (2021) also highlighted the effectiveness of microbial consortia, which can achieve recovery efficiencies above 90% for multiple metals. For plant design, bioleaching requires integration with mechanical shredding and the installation of bioreactors with aeration and pH control systems.

Direct regeneration offers a different philosophy by repairing or reactivating cathode powders without breaking them down into elemental metals. This method is energy-efficient and retains the crystal structure of the active material, making it particularly suitable for manufacturing scrap. However, it is highly sensitive to impurities and requires precise sorting of battery chemistries, which currently limits its industrial scalability (Yang et al., 2018; Roy et al., 2021).

The most forward-looking approaches combine these methods into hybrid or circular hydrometallurgical systems. Mechanochemical activation can be paired with hydrometallurgy to reduce acid consumption, while organic acid leaching can be combined with bio-assisted processes to further minimize environmental impact. Electrodialysis is increasingly being considered a complementary technology for selectively separating ions and reducing reagent use, aligning with the principles of a circular economy (Cerrillo-Gonzalez et al., 2024). These hybrid systems are designed to minimize waste, reuse reagents, and integrate renewable energy sources, making them attractive for future large-scale applications.

Safety and risk management are central to plant design. Fire, explosion, and thermal runaway remain major hazards during shredding, storage, and processing. Strategies to mitigate these risks include pre-discharging batteries in saline solutions, shredding under inert atmospheres, and designing modular facilities with fire suppression and gas scrubbing systems. Recent studies emphasize that integrating safety with sustainability is essential, as improper handling of LIB waste can lead to fires, toxic leachates, and environmental contamination (Kim & Park, 2025).

In comparative terms, pyrometallurgy is a mature and commercially deployed process, but it is also environmentally burdensome. Hydrometallurgy offers high recovery and purity, but it requires careful management of wastewater and the judicious use of chemicals to ensure optimal results. Bioleaching is emerging as a low-energy, eco-friendly option, though its slow kinetics remain a challenge. Direct regeneration is promising for certain feedstocks but remains industrially unviable. Hybrid and circular hydrometallurgical approaches represent the cutting edge, combining the strengths of different methods to achieve high recovery with reduced environmental impact.

Conclusively, the state of the art in LIB recycling plant design is shifting away from traditional pyrometallurgical dominance toward green, selective, and circular approaches. Hydrometallurgy remains the backbone, but bioleaching and organic acid leaching are gaining traction as sustainable alternatives. Mechanochemical activation and selective leaching enhance the efficiency of $LiFePO_4$. Electrodialysis and circular hydrometallurgy represent the next frontier for scalable, low-waste plants. Safety and regulatory integration (EPR, EU Battery Regulation 2023) are critical for industrial deployment.

Table 2.1: Comparative Decision Matrix for LIB Recycling Pathways

Pathway	Recovery Efficiency	CAPEX/OPEX	Environmental Impact	Scalability	Feedstock Suitability	Key References
Pyrometallurgy	High for Co/Ni/Cu; poor for Li/Al/Mn	High CAPEX (furnaces, gas cleaning); high OPEX (energy)	High GHG, toxic off-gases; slag disposal	Mature, industrial scale	Mixed chemistries, low sorting	Roy et al., 2021; Cerrillo-Gonzalez et al., 2024
Hydrometallurgy	>95% for Li/Co/Ni/Mn with optimized leaching	Moderate CAPEX; OPEX driven by reagents/water	Wastewater generation is mitigated by circular flows	Pilot to industrial	NMC, NCA, LCO; adaptable to LFP with pretreatment	Yang et al., 2017; Yang et al., 2018; Lerchbammer et al., 2023
Direct Regeneration	Variable; near-virgin performance possible	Low CAPEX/OPEX if the feedstock is clean	Low emissions; minimal reagents	Lab to pilot; scaling is limited by sorting	Manufacturing scrap; homogeneous EOL	Yang et al., 2018; Cerrillo-Gonzalez et al., 2024
Bio-hydrometallurgy	70–96% depending on pulp density and microbes	Low CAPEX; low OPEX (nutrients, aeration)	Very low toxicity; benign reagents	Pilot; kinetics limit throughput	Mixed black mass; low-grade feed	Roy et al., 2021; Garcia et al., 2025
Hybrid / Circular Hydro	High; combines strengths of hydro + bio/mechano	Moderate CAPEX; OPEX reduced by reagent recycling	Low if reagents regenerated; water loops closed	Emerging, promising for scale	Flexible, especially LFP and mixed	Cerrillo-Gonzalez et al., 2024; Akbar et al., 2025

Bioleaching has been selected as the core technology for this lithium-ion battery recycling plant due to its unique combination of environmental sustainability, high metal recovery efficiency, and low energy demand, making it ideal for modern circular economy applications.

Unlike pyrometallurgical processes, which require temperatures exceeding 1000 °C and generate toxic off-gases and slag (Roy et al., 2021; Cerrillo-Gonzalez et al., 2024), bioleaching operates under ambient temperature and pressure. This significantly reduces energy consumption and eliminates the need for complex gas-cleaning systems. This inherently safe, low-impact profile aligns with the plant's design philosophy and regulatory expectations for green industrial development.

Recent advances demonstrate that bioleaching can achieve rapid and high recovery rates. Garcia et al. (2025) reported that bio-oxidized Fe(III) leaching enabled 95% recovery of copper, nickel, and aluminum within 20 minutes, and over 90% recovery of cobalt, manganese, and lithium within 30 minutes. These results challenge the assumption that bioleaching is too slow and confirm its viability for high-throughput operations.

Moreover, bioleaching is compatible with mixed battery chemistries and does not require aggressive chemical reagents or extensive pretreatment. Roy et al. (2021) highlighted the effectiveness of microbial consortia in achieving recovery efficiencies above 90% across multiple metals, including lithium, cobalt, and manganese. This flexibility reduces the need for precise sorting, allowing the plant to process a diverse range of feedstocks, like end-of-life consumer batteries and manufacturing scrap.

From an environmental standpoint, bioleaching significantly reduces the consumption of harsh chemical reagents. While it still produces a wastewater stream, the chemical load is lower than in conventional hydrometallurgy. The microbial processes are self-sustaining and can be integrated with renewable energy sources, further reducing the plant's carbon footprint (Kim & Park, 2025). Operationally, bioleaching supports modular plant design, with bioreactors that can be scaled and controlled via aeration and pH monitoring. This modularity enhances safety, simplifies maintenance, and allows for phased capacity expansion.

Chapter 3. Market Analysis

3.1 Main Product: Lithium Carbonate (Li_2CO_3)

The main and most valuable product from this recycling plant is lithium carbonate. This form of lithium is the most common lithium compound used industrially. The battery sector dominates 70%+ of the global lithium carbonate demand. Additionally, the demand surge driven by EVs, along with expanding energy grid storage systems, will cause the battery sector to grow further (Bajolle et al., 2022). The global lithium carbonate market was valued at USD 28 million in 2024 and projected to reach USD 75 billion by 2030. This rapid growth reflects the large-scale increase in interest in lithium-ion batteries. The price of lithium carbonate by tonne has exhibited extreme volatility in recent years. From 2022 to 2024, the market showed a range from USD 10000 to a bubbled USD 80000. This overall volatility is attributed to uneven production capacity expansions, EV adoption rates, speculative trading, and geopolitical supply risks in key producing countries (Bajolle et al., 2022).

3.2 Secondary Product: Cobalt Phosphinate ($Co(H_2PO_2)_2$

The cobalt market in 2024 experienced large shifts, with global demand surpassing 200,000 tonnes for the first time and battery applications accounting for 76% of all cobalt consumed. Demand from EVs alone comprised 43% of cobalt usage, while superalloys and military applications remained important non-battery sectors. Regional issues spiking oversupply from the Democratic Republic of Congo and from Indonesia resulted in a sharp decline in cobalt prices. Supply from secondary sources, mainly recycled lithium-ion batteries and production scrap, grew by 13% to reach 22,000 tonnes, representing 8% of total cobalt supply. However, in percentage terms, recycling lagged behind primary supply (Cobalt Institute, 2025). Pricing is tied to global metal spot prices

3.3 Global Market

Within the lithium supply chain, lithium carbonate plays a critical role. Primarily used in lithium-ion batteries, lithium carbonate is also used in other industrial applications.

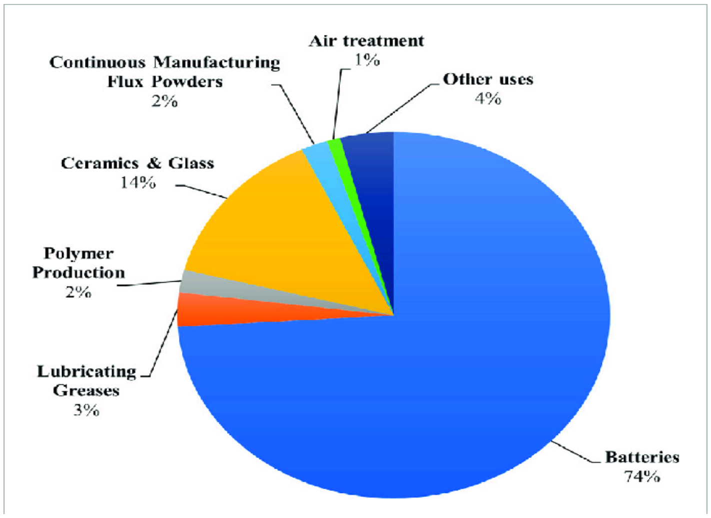

FIGURE 3.3: 1 PIE CHART SHOWING MARKET END USES OF LITHIUM AS A PERCENTAGE OF GLOBAL CONSUMPTION IN 2021 (U.S GEOLOGICAL SURVEY, 2022).

The global market capacity for lithium carbonate is shaped by the lithium resource base, mining and production (including recycling) capacity, and the growing electric vehicle market. The mineral lithium is typically extracted from one of two deposits, brines or ores. In South America, Chile, Argentina and Bolivia contain substantial amounts of lithium carbonate in their brine deposit. Using evaporation methods, lithium carbonate is readily produced. Pegmatite ore deposits are significant in Australia, Canada, parts of Africa and China. Future supply diversification can be seen in lithium-bearing clays such as hectorite. Geothermal or oilfield brines, which are less significant presently, have tremendous potential with extraction research. Additionally, lithium carbonate chemical production is closely tied to conversion facilities, which process lithium concentrates/brines into usable battery-grade lithium carbonate and similar derivatives (Martina et al, 2017).

3.4 Market Trend

From 2020 to 2025, the post-COVID electric vehicle caused unusual pricing. Mining output surged, with production reaching historic highs globally. However, this supply growth would soon outpace demand, now leading to oversupply in some sectors, subsequently causing immense price volatility. The market in the few years ahead is set to tighten further until the supply surplus is chewed through. However, from 2035 onwards, lithium usage is expected to see another large growth with electrical grid storage systems requiring expansion (Calisaya-Azpilcueta et al., 2020).

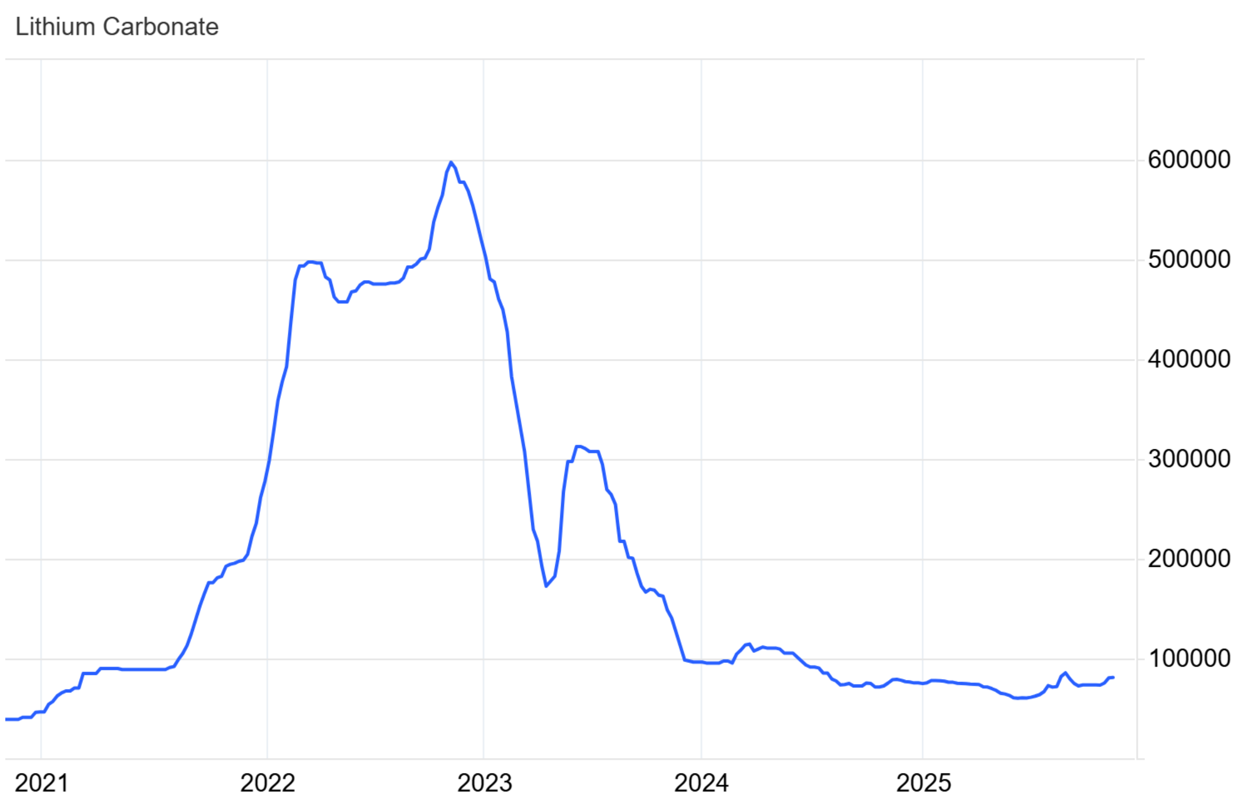

FIGURE 3.4 LITHIUM CARBONATE PRICING (CNY/TONNE) (TRADING ECONOMICS)

3.5 Global Scale

Globally, as lithium-ion batteries are integrated into local infrastructure, including energy grids and EVs, the market, including recycling, is projected to continue to grow. Regionally, China is the largest consumer. This is driven by record EV production and rapid deployment of electric grid storage systems. China controls significant portions of the lithium supply chain from raw material procurement straight to cell production and end product assembly. Building this recycling plant within China allows us to leverage proximity to key buyers in the industry. This results in a reduction in the need for logistics and general trade complexities. This local presence not only ensures streamlined access to one of the world's fastest-growing lithium markets but also aligns with China's strategic industrial policies promoting domestic supply chain integration. As a result, our plant will capitalize on China's role as a market leader, enhancing our competitive advantage through both operational efficiency and closer engagement with the global lithium market's most influential buyer base (Zehong et al, 2024).

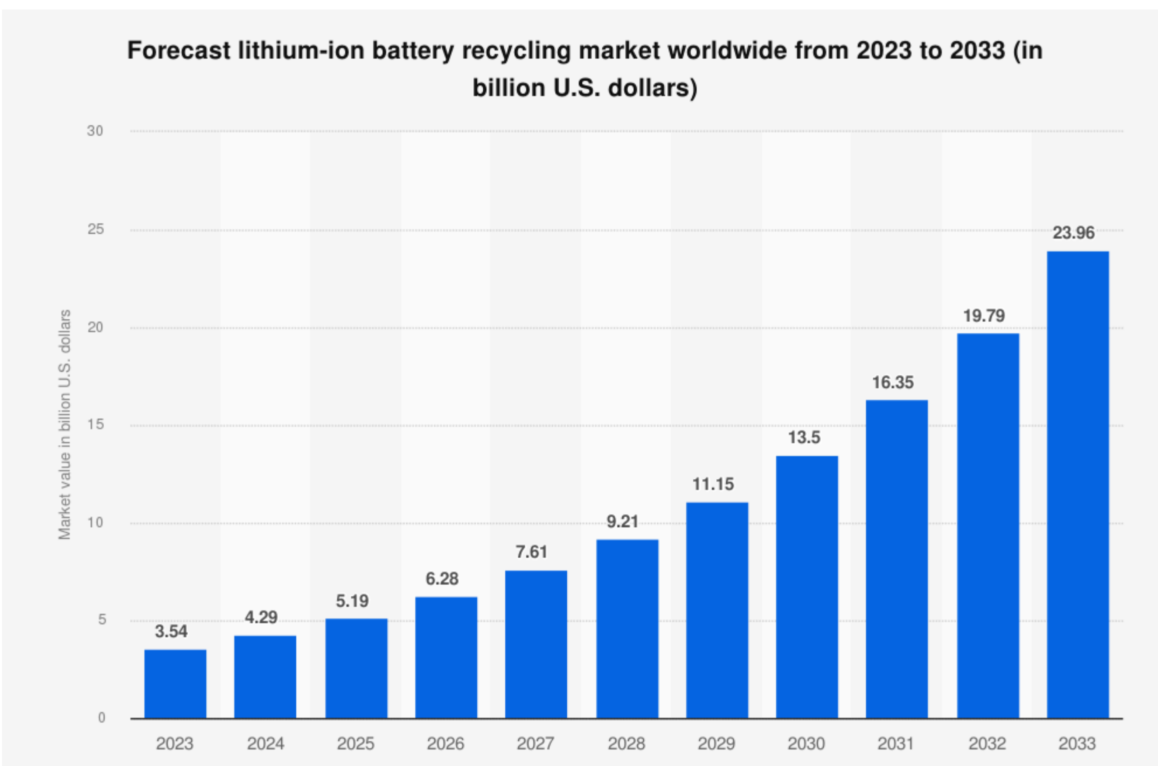

FIGURE 3.5 LIB BATTERY RECYCLING FORECAST (BIS RESEARCH, 2025)

3.6 Raw Material: Used Lithium Ion Batteries

The lithium-ion battery depends on various key raw materials, with lithium, cobalt, nickel, and graphite being the main components. Lithium primarily comes from brine deposits in South America, specifically Chile, Argentina, and Bolivia, and hard rock deposits of spodumene mainly found in Australia. Cobalt, which is crucial for battery cathodes, is mined mostly in the Democratic Republic of Congo, while nickel and graphite supplies are from various parts of the world that include Indonesia, Russia, and China. Our strategically located plant within China, therefore, puts us near one of the biggest lithium-ion battery manufacturing hubs in the world, with significant graphite reserves and increasing cobalt processing facilities. The market trend is a high demand for all these raw materials as electric vehicles and grid storage technologies expand rapidly. China is leading both in consumption and processing, commanding a huge portion of the battery supply chain downstream. With this in place, China can highly impact raw material demand and the pricing dynamics, and integration into this ecosystem allows us to optimize the logistics within the supply chain and effectively capitalize on market growth.

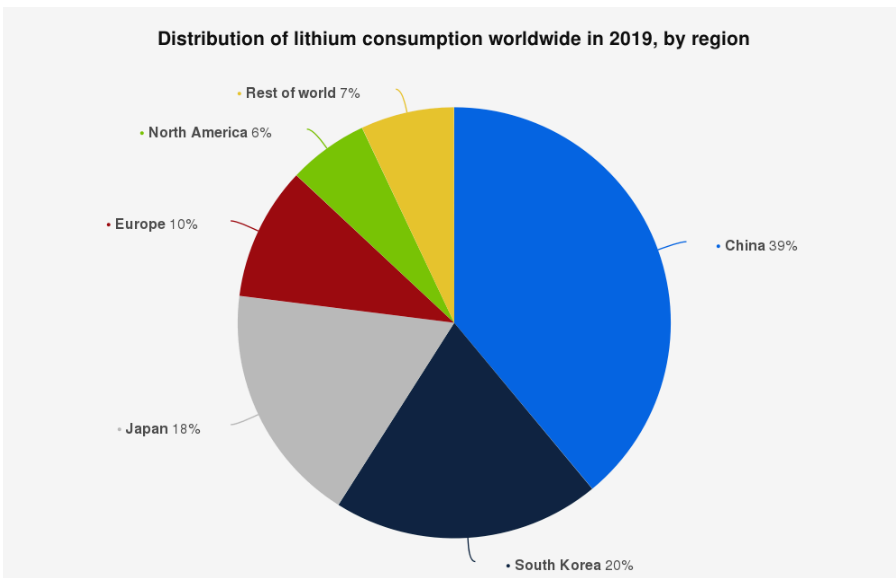

FIGURE 3.6 LIB BATTERY RECYCLING FORECAST (BIS RESEARCH, 2025)

Recycling spent LIBs offers significant cost benefits by reducing raw material expenses by approximately 13-57%, conserving natural resources, and mitigating environmental impacts. Despite technological advances, pricing is also affected by the complexity and cost of recycling processes, including energy consumption and reagent use. (Zhaocheng, et al. 2025).

3.7 Profitability

The profitability of our recycling plant is dependent on the recovery of the precious metals. As per the mass balance, expected flow rates of process materials can be interpreted to determine monthly profits per metal. Expectedly, lithium carbonate recovery yields the most profit and will essentially dictate the capability of the plant. The other metals account for only 6% of projected profits. Expense-wise,

Table: 3.7 Profitability of product obtained estimates

Component	kg/h	monthly tonne	usd/tn (price)	Profit
Lithium Carbonate	270.8	194.94	578571	$112,787,050.12
Cobalt Phosphinate	41.4	29.81	48570	$1,447,774.56
Nickel Hydroxide	380.2	273.74	15102	$4,134,050.63
Copper 2+	215.9	155.44	11134	$1,730,656.19

Chapter 4. Process and Equipment Selection

4.1 Process Selection

The selection of an appropriate process technology is fundamental to achieving the technical, economic, and environmental objectives of the proposed hydrometallurgical battery recycling facility. The chosen process must ensure high metal recovery efficiency, low waste generation, and compatibility with environmental regulations. From the technological survey completed earlier, the best choice is bioleaching through biohydrometallurgy using A. Ferrooxidans.

Bioleaching stands out as a recycling pathway because it combines environmental responsibility with technical effectiveness, but it also comes with limitations that must be acknowledged. At its core, the process relies on microorganisms such as A. Ferrooxidans to oxidize ferrous iron or sulfur compounds, generating ferric ions and acids that dissolve the valuable metals from spent lithium-ion batteries. This biological approach operates at ambient temperature and pressure, which means it avoids the extreme energy demands of pyrometallurgy and the heavy chemical consumption of conventional hydrometallurgy. As a result, bioleaching is inherently safer, produces fewer toxic emissions, and requires less capital investment in high-temperature equipment. It is also flexible, as microbial consortia can adapt to mixed and variable feedstocks. Recent studies have shown that recovery rates above 90% are achievable for lithium, cobalt, nickel, manganese, copper, and aluminum. The modular nature of bioreactors enables the process to be scaled, allowing facilities to expand capacity in phases while maintaining control over aeration and pH levels.

Despite these advantages, bioleaching is not without drawbacks. The kinetics of microbial reactions are slower than purely chemical leaching, which can limit throughput unless reactors are carefully optimized. The process is sensitive to operating conditions, as pH, temperature, oxygen transfer, and pulp density must be tightly controlled. Additionally, the presence of toxic electrolytes can inhibit microbial activity. While the wastewater generated is less chemically loaded than in hydrometallurgy, it still requires treatment before discharge.

Another challenge is that, compared to pyrometallurgy and hydrometallurgy, bioleaching is less mature at full industrial scale, meaning that large-scale deployment still carries some uncertainty. In short, bioleaching offers a green, low-energy, and increasingly viable solution for lithium-ion battery recycling, but its slower kinetics and operational sensitivities remain important considerations for plant design.

The table below summarizes how effective each method is by ranking them from 1 to 5 on each criterion, with lower numbers indicating poor performance and higher scores indicating excellent performance.

TABLE 4.1.1: A SIMPLIFIED PROCESS FLOW DIAGRAM OF THE SELECTED PROCESS

Criteria	Pyrometallurgy	Hydrometallurgy	Bioleaching	Direct Regeneration	Hybrid Hydro
Cost Effectiveness (CAPEX/OPEX)	2	3	4	4	3
Product Quality (Purity, Recovery)	3	5	4	5	5
Safety	2	3	5	4	4
Environmental Impact	1	3	5	5	5
Scalability	5	4	3	2	3
Feedstock Flexibility	5	4	4	2	5
Technology Maturity	5	4	3	2	2
Total	**23**	**26**	**30**	**22**	**27**

Through this rating table, our selection of bioleaching was previously supported by a tabulated expression and result, as shown in Tables 4.1.1 and 2.1.

A simplified process flow diagram (PFD) can be illustrated to demonstrate the idea behind our plant design, detailed later in Chapter 8. As seen in Figure 4.1.1, we have laid out the general purpose of bio-hydrometalurgy through bioleaching of batteries.

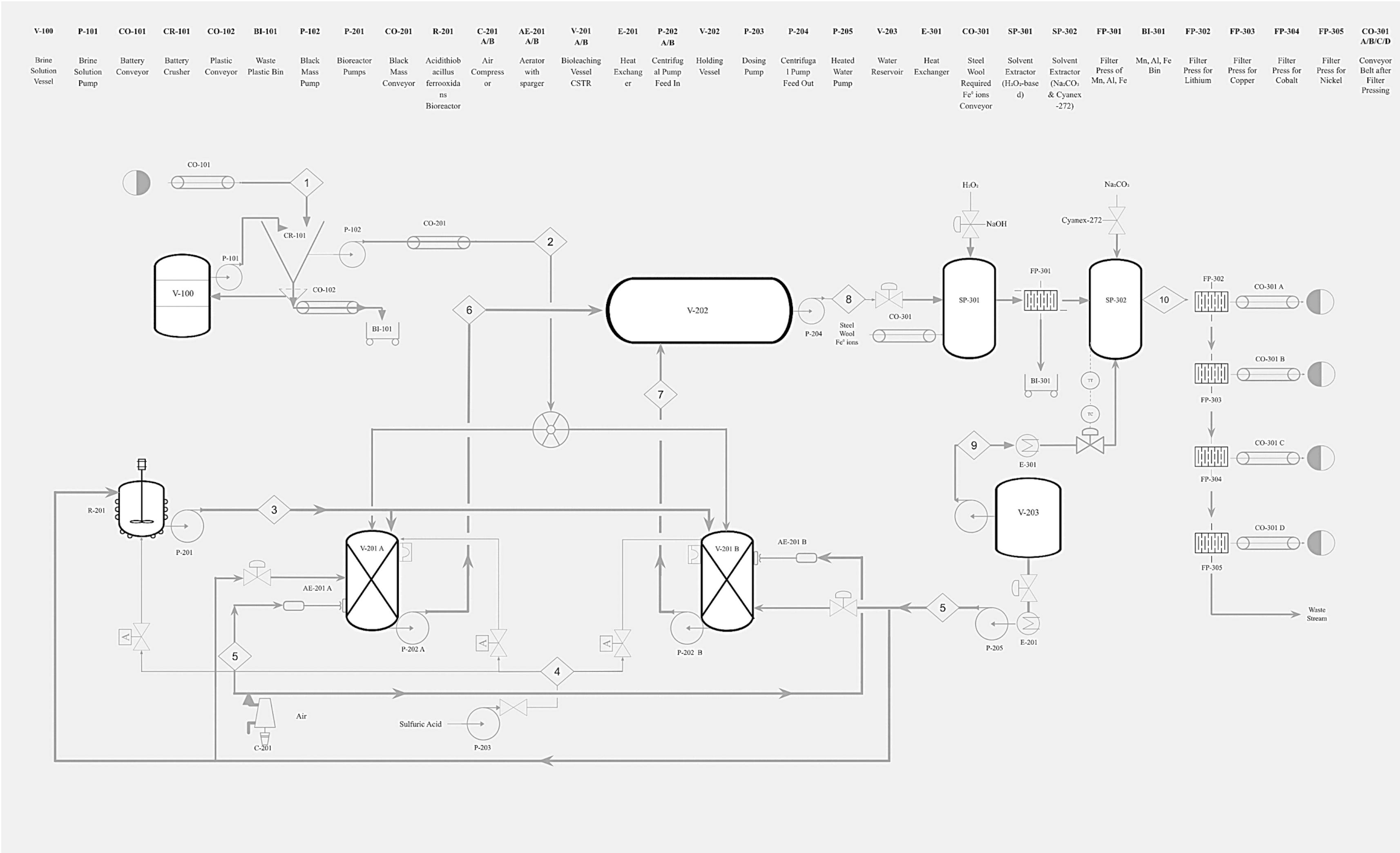

FIGURE 4.1.1: A SIMPLIFIED PROCESS FLOW DIAGRAM OF THE SELECTED PROCESS

Zone 1: Pretreatment

The desired input of our plant design is 15000 tons of unsorted batteries every year to be processed to obtain Lithium Carbonate. The batteries are submerged and crushed in a wet brine solution consisting of Sodium Chloride to ensure electrical discharge and prevention of flare-up due to inert conditions. The crusher will have a major waste stream of all plastics and other waste that float to the top of the solution as crushing and vibrations occur. The desired output of the crusher is the unsorted black mass of metals consisting of the desired metals, Li, Co, Ni, Cu.

Zone 2: Leaching Zone

After the batteries are broken down into black mass, it is transported into the CSTR to begin bioleaching. For this to occur, a culture medium must be added from the Acidithiobacillus Ferrooxidans Bioreactor in a 1 (black mass):100 (culture medium) ratio. It is also important to keep the bioreactor and the CSTR at 27°C, which is done using a heat exchanger and water, as well as 1.5 PH, which is maintained by H2SO4. After the bioleaching is complete, we are left with a solution that is transported to a large holding vessel waiting to be extracted in zone 3. The solution in the holding vessel will have all desired metals inside of it in elemental form.

Zone 3: Recovery Zones

Using properties of PH and the ability to separate out certain metals depending on PH level, we are able to obtain our desired metals. Co, Li, Cu, and Ni are obtained through a filter press after PH levels have been managed at differing reacting times to ensure high metal recovery. Cu, Al, Mn, and Fe(III) are extracted in one separator, and Co, Li, and Ni are extracted together in another separator. The first step is the Cu extraction. This is done by adding Fe^0 using steel wool. This will precipitate out the Cu. Mn, Al, and Fe(III), which are impurities that are extracted by simply raising the PH. The solution now leaves the first separator into a filter press, where all the impurities are extracted. The solution now enters a second separator, where the rest of our desired metals are extracted. Co extraction, which involves adding Cyanex-272. This reaction is at equilibrium, favouring the production of cobalt phosphinate. Once equilibrium is achieved, the PH is then increased to 10 by adding NaOH. Ni will automatically precipitate as it turns to $Ni(OH)_2$. To extract the Li, Na_2CO_3 is added, where it becomes Li_2CO_3. The reason Li is extracted at the end is that the recovery of Li is the main objective, and removing all the metals and impurities first will boost the recovery rate of Li.

4.2 Equipment Selection

Bioleaching is the selected core technology due to its low energy demand, modular scalability, and high recovery potential when coupled with downstream hydrometallurgy. The layout below identifies critical equipment, compares options, and recommends configurations tailored to an ambient-temperature, aerated, pH-controlled plant handling mixed black mass feeds.

4.2.1 Front-end solids preparation

Crusher

The crusher is a critical component in the front-end preparation of black mass, ensuring that coarse fragments are reduced to a controlled particle size suitable for slurry preparation and downstream bioleaching. In this process, the crusher must operate with a submerged input solution, as the goal is to discharge batteries into a brine solution. Meaning the feed enters already wetted or within a slurry environment. This design eliminates dust emissions, enhances safety during the handling of hazardous fines, and integrates seamlessly with slurry pumping and wet screening.

Several equipment options are available for submerged or wet crushing. Wet hammer mills can handle mixed feed and deliver high throughput, but they tend to generate excessive fines and suffer from accelerated wear in abrasive slurries. Inline twin-shaft macerators or shredders are fully enclosed and operate directly in slurry lines, producing consistent coarse reduction while eliminating dust and aerosols. However, they are limited in throughput and produce a coarser product than mills. Chopper pumps combine pumping and chopping in a compact unit, but their size reduction capability is limited by impeller geometry, and they consume more energy per tonne. Wet attrition mills, also known as rotor-stator disintegrators, offer narrower particle size distributions and are suitable for secondary refinement, although they require pre-reduction to prevent plugging. Finally, wet ball or SAG mills offer excellent control down to sub-millimetre sizes, but are capital-intensive and best suited as secondary or tertiary stages rather than primary submerged crushers.

However, our recommendation, which is also the most appropriate configuration, is an inline twin-shaft macerator as the primary submerged crusher, followed by a wet attrition mill for secondary refinement if tighter size control is required. This arrangement ensures enclosed, dust-free operation, integrates seamlessly with slurry pumping, and delivers a particle size distribution that balances slurry stability with downstream oxygen transfer requirements. The flow path involves a submerged feed hopper discharging through an isolation gate into the macerator, then into a rubber-lined centrifugal slurry pump. This is followed by optional wet screening or attrition milling before feeding the conditioning CSTRs. This configuration minimizes dust, ensures safe handling of hazardous fines, and provides reliable slurry preparation for bioleaching.

TABLE 4.2: PROS & CONS OF CRUSHER OPTIONS

Crusher Type	Advantages	Disadvantages
Wet Hammer Mill	High throughput; Tolerant to mixed feed; Simple internals	Generates excess fines; High wear in abrasive slurry; Requires robust sealing
Inline Twin-Shaft Macerator	Fully enclosed, submerged operation; Eliminates dust and aerosols; Consistent coarse reduction; Easy isolation with valves	Coarser product than mills; Limited throughput per unit; Needs a magnet trap to protect cutters
Chopper Pump	Combines pumping and chopping; Compact footprint; Handles fibrous/irregular feed	Limited size reduction fidelity; High energy per tonne; Cutter/impeller wear in abrasive slurry
Wet Attrition Mill	Narrower particle size distribution; Adjustable shear; Good for secondary refinement	Requires pre-reduction; Sensitive to very coarse fragments; Risk of plugging

Wet Ball/SAG Mill	Excellent size control (sub‑mm); Scalable for high throughput; Produces uniform slurry	High CAPEX and OPEX; Not suitable as primary submerged crusher; Requires pre‑sizing

Conveyor

Conveyors transfer solids between crushing, screening, and slurry preparation. Belt conveyors are efficient for high throughput but require dust control, while screw conveyors are better for short, enclosed transfers. Tubular drag conveyors offer excellent dust containment, but they are slower and more expensive.

Our recommendation is that an enclosed belt conveyor with dust hoods is selected for the main transfer, supplemented by short screw conveyors for slurry tank feeding. This is justified by the fact that enclosed conveying reduces dust and improves safety in handling hazardous battery waste (BRGM, 2021).

Table 4.3: Pros & Cons of Conveyor (Enclosed Belt + Screw)

Advantages	Disadvantages
Belt conveyors handle high throughput efficiently (Metso, 2023)	Require dust control and spill management
Gentle handling reduces particle degradation	Open belts can release dust if not enclosed
Enclosed design improves safety and environmental control (BRGM, 2021)	Higher CAPEX for enclosed or tubular systems
Screw conveyors are compact and good for short, steep transfers	Cause wear, higher power draw, and shear on particles

4.2.2 Liquid handling and reactor feed

Pump

Pumps move acidic slurry between unit operations. Rubber-lined centrifugal pumps are reliable for continuous slurry transfer, while peristaltic pumps provide precise dosing for reagents. Diaphragm pumps are versatile but less efficient than other types of pumps.

Our recommendation is to use rubber-lined centrifugal pumps, which are suitable for slurry transfer, and peristaltic pumps for reagent dosing. This is justified by the fact that slurry pumps with corrosion-resistant linings are standard in hydrometallurgy to handle abrasive and acidic streams (Metso, 2023).

Table 4.4: Pros & Cons of Pump (Centrifugal + Peristaltic)

Advantages	Disadvantages
Rubber-lined centrifugal pumps handle abrasive, acidic slurry (Metso, 2023)	Efficiency drops with viscous or high-solids slurries
Low maintenance and reliable for continuous duty	Limited head compared to positive displacement pumps
Peristaltic pumps provide precise dosing of reagents	Hose wear and lower flow capacity
Corrosion-resistant materials extend service life	Diaphragm pumps introduce pulsation and lower efficiency

4.2.3 Biochemical conversion

CSTR (conditioning)

Conditioning tanks homogenize slurry and stabilize pH before bioleaching. A single CSTR is simple but less flexible, whereas two in series allow for staged reagent addition and a better residence time distribution.

We recommend selecting two CSTRs in series for conditioning. This is because stirred tank systems are widely used in bioleaching due to their high mixing efficiency and scalability (Neale, 2010).

Table 4.5: Pros & Cons of CSTR (Conditioning)

Advantages	Disadvantages
Provides good mixing and homogenization (Neale, 2010)	Higher CAPEX for multiple tanks in series
Allows staged reagent addition (acid, Fe^{2+}, inoculum)	Requires more instrumentation and control
Improves residence time distribution	Larger footprint compared to a single tank
Stabilizes feed to bioleach reactors	More complex piping and operation

Bioreactor

The bioleaching core requires reactors that support microbial oxidation of Fe^{2+}. Stirred tank bioreactors handle solids well and allow precise control, though they consume more energy. Airlift reactors are efficient for oxygen transfer but less suited to slurries. Packed beds are prone to clogging.

We recommend using mechanically agitated CSTR-type bioreactors in series for slurry bioleaching, with an airlift reactor for ferric regeneration on clarified liquor. This is because commercial bioleaching plants use mechanically agitated reactors for slurry systems, with airlift reactors applied to clarified streams for efficient Fe^{2+} oxidation (Rawlings & Johnson, 2007; BRGM, 2021).

Table 4.6: Pros & Cons of Bioreactor (Stirred CSTRs + Airlift Loop)

Advantages	Disadvantages
Stirred tanks handle 5–10% solids effectively (Rawlings & Johnson, 2007)	Higher energy consumption for agitation
Precise control of pH, ORP, and DO (Neale, 2010)	Impeller wear in abrasive slurry
Modular and scalable in series	More complex operation and monitoring

Airlift loop provides efficient Fe^{2+} oxidation in clarified liquor (BRGM, 2021)	Not suitable for high-solids slurry; requires side-stream clarification

4.2.4 Thermal management

Heat exchanger

Temperature control is modest but necessary to maintain microbial activity. Shell-and-tube exchangers are robust but bulky, plate-and-frame exchangers are compact and efficient for clean streams, and jackets/coils are best suited for slurry reactors.

We recommend using Plate-and-frame exchangers for clarified liquor and jacketed reactors for slurry, as maintaining microbial optimum temperatures (25–35 °C) is critical for bioleaching kinetics (BRGM, 2021).

Table 4.7: Pros & Cons of Heat Exchanger (Plate-and-Frame + Jackets)

Advantages	Disadvantages
Plate-and-frame exchangers are compact, efficient, and easy to clean (Metso, 2023)	Gaskets must be chemically compatible; not suited for abrasive slurry
Jackets or coils avoid fouling in slurry reactors	Limited heat transfer area compared to dedicated exchangers
Counter-current flow provides good thermal efficiency	Shell-and-tube alternatives are bulkier and more expensive
Simple temperature trimming for modest exotherm	Additional cooling water system is required

4.2.5 Batch operations and downstream separations

Batch reactor

Batch reactors are used for precipitation and oxidation steps. Glass-lined steel offers superior corrosion resistance but is costly, whereas rubber-lined steel is more economical and sufficient for ambient acidic conditions.

We recommend using rubber-lined steel batch reactors for precipitation and oxidation processes. This is because hydrometallurgical precipitation steps are commonly carried out in lined steel tanks due to their chemical resistance and cost-effectiveness (Metso, 2023).

Table 4.8: Pros & Cons of Batch Reactor (Rubber-Lined)

Advantages	Disadvantages
Cost-effective for acidic, ambient-temperature service (Metso, 2023)	Less durable than glass-lined steel
Resistant to corrosion from acidic solutions	Limited temperature tolerance
Suitable for precipitation and oxidation steps	Requires agitation and hold time, increasing batch cycle length
Easy integration with pH and oxidant dosing	Mechanical lining damage risk

Filter press

Solid–liquid separation is critical for recovering precipitates and clarifying leachate. Plate-and-frame presses are flexible but labour-intensive, membrane presses produce drier cakes, and vacuum belt filters are continuous but yield wetter cakes.

In this case, a membrane filter press is selected for a drier cake and cleaner filtrate. This is supported by the fact that filter presses are standard in hydrometallurgy for producing high-purity filtrates and minimizing moisture in residues (Neale, 2010).

Table 4.9: Pros & Cons of Filter Press (Membrane)

Advantages	Disadvantages
Produces drier cake than plate-and-frame (Neale, 2010)	Higher CAPEX
Cleaner filtrate for downstream recovery	Batch operation, not continuous
Handles high solids loading effectively	Requires labour for cloth maintenance and cycle management

Wash water can be recycled to minimize waste	Larger footprint than some continuous filters

This equipment set strikes a balance between cost, control, and safety for a bioleaching-based LIB recycling plant. The crusher and enclosed belt conveyors deliver predictable particle size and clean handling. Rubber-lined centrifugal pumps and staged conditioning CSTRs stabilize slurry feed. Stirred bioleach CSTRs, backed by an airlift ferric loop, maximize oxygen transfer and ORP while tolerating 5–10% solids. Plate-and-frame exchangers and reactor jackets manage modest heat loads without fouling. Rubber-lined batch reactors and membrane filter presses produce clean filtrates and dry cakes for high-purity downstream recovery. The configurations and flow paths chosen reflect the process's ambient operation, corrosion environment, solids content, and the need for modular, scalable control.

Chapter 5. Material Properties

This chapter outlines the materials involved in the selected process, including raw materials, products, and by-products. A clear understanding of the chemical and physical properties of these substances is essential for safe plant design, process optimization, and regulatory compliance.

Section 5.0 identifies all materials used or generated in the process and summarizes their functions. Section 5.0.1 provides a concise overview of their important properties, highlighting corrosivity, reactivity, toxicity, and other safety-critical characteristics. To support safe operations, 5.0.2 directs the reader to the Material Safety Data Sheets (MSDS) provided in the appendix, which contain detailed hazard, handling, and emergency response information for each material.

Section 5.1 focuses specifically on hazardous materials. It identifies substances classified as hazardous under WHMIS/GHS and other applicable regulatory frameworks, outlines threshold concentrations, and specifies appropriate storage, handling, and protective measures. This section also details the required personal protective equipment (PPE), safety devices, and operational protocols to ensure worker safety and environmental protection.

Together, these sections provide a comprehensive foundation for understanding the chemical risks associated with the process and demonstrate how safety considerations are integrated into the overall plant design.

TABLE 5.0.1: MATERIALS INVOLVED IN THE PROCESS

Category	Material	Function / Notes
	Sulfuric acid (H_2SO_4)	Leaching agent
	Spent secondary Batteries	Primary feedstock
	Processing Water	Universal solvent

Raw Material	Ferric sulfate ($Fe_2(SO_4)_3$) / Ferrous sulfate ($FeSO_4$)	Oxidants/mediators
	Hydrogen peroxide (H_2O_2, 3–35%)	Oxidant for Mn precipitation
	Biogenic acids (acetic, gluconic, microbial acids)	Microbial leaching aids
	Sodium hydroxide (NaOH)	Base for precipitation
	Sodium carbonate (Na_2CO_3)	Base for Li_2CO_3 precipitation
	Solvent extraction reagents (DEHPA, Cyanex 272 in kerosene)	Cobalt separation
	Sodium chloride brine (NaCl solution)	Neutralization/discharge medium
	Nitrogen/argon gases	Inert shredding atmosphere
	Microbial media (6K medium)	Nutrient solution for bioleaching
Product	Copper (Cu)	Final recovered metal
	Aluminum hydroxide ($Al(OH)_3$)	Final recovered salt
	Manganese dioxide (MnO_2)	Final recovered oxide
	Cobalt hydroxide ($Co(OH)_2$)	Final recovered hydroxide
	Nickel hydroxide ($Ni(OH)_2$)	Final recovered hydroxide
	Lithium carbonate (Li_2CO_3)	Final recovered carbonate
By-product / Intermediate	Sodium sulfate (Na_2SO_4)	Neutralization product
	Aqueous raffinate streams	Residual salts in solution
	Off-gases (SO_x, O_2, CO_2, HF)	Gaseous emissions
	Graphite/Binder Sludge	Solid waste product

	Plastics and Metal Scrap	Non-hazardous solid waste

Table 5.0.2 outlines the MSDS requirements for each material, noting that full MSDS-style summaries are provided in the Appendix and that official supplier SDSs must be maintained on file for WHMIS/GHS compliance purposes.

TABLE 5.0.2: SUMMARY OF IMPORTANT PROPERTIES

Material	Key Properties
Spent Lithium-Ion Batteries	Primary feedstock. Hazardous: contains flammable electrolytes, toxic metals, and stored energy.
Sulfuric acid (H_2SO_4)	Strong mineral acid; highly corrosive; reacts exothermically with water
Ferric sulfate ($Fe_2(SO_4)_3$) / Ferrous sulfate ($FeSO_4$)	Iron salts; irritant dusts; environmental hazard in large releases
Hydrogen peroxide (H_2O_2, 3–35%)	Strong oxidizer; unstable with contaminants; causes burns
Sodium hydroxide (NaOH)	Strong caustic; severe burns; exothermic dissolution
Sodium carbonate (Na_2CO_3)	Mild base; irritant; CO_2 release on neutralization
Scrap Iron (Fe)	Solid reductant for copper cementation
Solvent extraction reagents (DEHPA, Cyanex 272 in kerosene)	Flammable organic liquids; irritant; environmental hazard
Sodium chloride brine (NaCl solution)	Low hazard; electrical safety concern in live systems
Nitrogen/argon gases	Asphyxiation risk in confined spaces; pressure hazards
Microbial media (6K medium)	Low hazard; BSL-1 organisms; aerosols may irritate

Copper (Cu) powder	Final product. Inhalation hazard.
Metal product salts (Li_2CO_3, $Co(OH)_2$, $Ni(OH)_2$, $MnCO_3$, $FePO_4 \cdot 2H_2O$)	Dust inhalation hazard; Ni/Co compounds are sensitizers/carcinogens
Graphite/Binder Sludge	Graphite/Binder Sludge
Solid Waste ($Al(OH)_3$, $Fe(OH)_3$, MnO_2)	Waste precipitates. Contains aluminum, iron, and manganese.
Sodium Sulfate (Na_2SO_4)	Main salt byproduct in final wastewater stream.
Off-Gases (HF, O_2, CO_2)	HF (highly toxic gas) from crusher; O_2/CO_2 from bioreactors.

5.1 Hazardous Materials

Table 5.1.1 identifies the hazardous materials used in the process, their corresponding regulatory classifications, and the storage, handling, and protective equipment requirements necessary to ensure safe plant operation.

Table 5.1.1: Hazardous Materials and Safety Measures

Hazardous Material	Regulatory / Classification	Storage & Handling	Protective Equipment / Safety Devices
Sulfuric acid (H_2SO_4)	WHMIS: Corrosive (Skin Corr. 1A, H314); CEPA listed	Segregated acid storage; corrosion-resistant containment	Acid-resistant gloves, face shield, apron, fume hood
Hydrogen peroxide (H_2O_2 >8%)	Oxidizing Liquid Cat. 1–3 (GHS); controlled storage thresholds	Vented, temperature-controlled storage; segregate from organics/metals	Goggles, face shield, gloves; avoid contamination

Sodium hydroxide (NaOH)	WHMIS: Corrosive (Skin Corr. 1A, H314); threshold ≥0.5%	Dry storage; segregate from acids, ammonium salts, and aluminum	Alkali-resistant gloves, goggles/face shield
Solvent extraction reagents (DEHPA, Cyanex 272, kerosene)	Flammable Liquid Cat. 3; Irritant; Environmental Hazard	Flammable-liquid cabinets; ground/bond containers; explosion-proof ventilation	Solvent-resistant gloves, goggles, flame-proof lab coat
Nickel hydroxide ($Ni(OH)_2$)	Carcinogenicity Cat. 1B (IARC Group 1); Sensitizer	Sealed containers; minimize dust; local exhaust	Respirator (N95+), gloves, goggles; closed handling
Cobalt hydroxide ($Co(OH)_2$)	Carcinogenicity Cat. 1B; Sensitizer	Sealed containers; minimize dust; local exhaust	Respirator (N95+), gloves, goggles; closed handling
Nitrogen / Argon gases	Simple asphyxiants; Compressed gas regulations	Cylinders secured upright; O_2 monitoring in enclosed spaces	Cylinder restraints, regulators, O_2 monitors

Chapter 6. Reaction Kinetics and Empirical Rules

This plant's design does not involve conventional chemical reactions with high temperatures or pressures. Instead, it operates as a biohydrometallurgical process, relying on biologically-catalyzed and synergistic chemical reactions to dissolve the metals from the solid black mass feedstock. The process utilizes a biogenic leaching agent, specifically ferric iron Fe(III), produced through the bio-oxidation of ferrous iron Fe(II) by the bacterium Acidithiobacillus Ferrooxidans.

Bio-Leaching

The use of a biogenic leaching agent, specifically ferric iron (Fe^{3+}), produced via A. Ferrooxidans bio-oxidation is evaluated while maintaining a 1% weight-to-volume ratio in the solution (Garcia et al., 2025). The primary biological reaction, shown in Equation 6-1, is the continuous regeneration of the Fe^{3+} lixiviant:

$$4\,Fe^{2+} + 4H^{+} + O_2 \rightarrow 4Fe^{3+} + 2H_2O\ \textit{biological activity} \qquad \text{(Eq. 6-1)}$$

For the transition-metal oxides found in cathode phases (e.g., $LiNiMnCoO_2$, $LiCoO_2$), dissolution is accelerated by reductants that lower the oxidation state, as the resulting M^{2+} species are more soluble than their M^{3+} or M^{4+} forms. In mixed systems that contain elemental Cu and Al fragments, the biogenic ferric iron (Fe^{3+}) acts as a strong oxidant for these elemental metals (Eq. 6-2 and 6-3). This, in turn, generates ferrous iron (Fe^{2+}) in situ:

Reductive assistance (Fe(III)/Fe(II), H2O2, organics)

In mixed systems that contain elemental Cu/Al (foils or fragments), ferric iron acts as a strong oxidant for elemental metals, while the generated ferrous iron can reduce high-valent transition metals in oxides, accelerating dissolution:

$$Cu^{0} + 2Fe^{3+} \rightarrow Cu^{2+} + 2Fe^{2+} \qquad \text{(Eq. 6-2)}$$

$$Al^{0} + 3Fe^{3+} \rightarrow Al^{3+} + 3Fe^{2+} \qquad \text{(Eq. 6-3)}$$

Proton‑attack dissolution of metal oxides (acid leaching):

$$MO + 2H^{+} \rightarrow M^{2+} + H_2O \qquad \text{(Eq. 6-4)}$$

Transition‑metal oxides in cathodes dissolve faster when reduced to lower valence states. Reductants such as Fe(II) or H_2O_2 accelerate dissolution by converting $Co^{3+}/Ni^{3+}/Mn^{4+}$ to soluble $Co^{2+}/Ni^{2+}/Mn^{2+}$ (Yang et al., 2018; Lerchbammer et al., 2023).

Redox‑coupled acceleration: Elemental Cu and Al react rapidly with Fe^{3+}, producing Fe^{2+} that reduces transition‑metal oxides. This explains the very fast kinetics observed in unsorted black mass bio-oxidized with Fe(III): ~100% Cu/Al in 20 minutes, and over 90% Ni/Co/Li in ~30 minutes (Garcia et al., 2025).

Separation & Recovery Chemical Reactions

This process begins with the Pregnant Leach Solution (PLS) from the bioleaching circuit, which contains all the dissolved metals as sulfates (e.g., $CuSO_4$, $Al_2(SO_4)^3$, $MnSO_4$, $CoSO_4$, $NiSO_4$, and Li_2SO_4).

Step 1: Copper (Cu) Recovery (Cementation)

Copper is first selectively removed from the PLS via cementation, where scrap iron (Fe) is used as a sacrificial reductant. The more noble copper precipitates as a solid metal, while the iron dissolves.

$$CuSO_{4(aq)} + Fe_{(s)} \rightarrow FeSO_{4(aq)} + Cu_{(s)} \qquad \text{(Eq. 6-5)}$$

Step 2: Aluminum (Al) Precipitation

After copper removal, the solution pH is raised to a range of 4.5-5.5 using a base, such as sodium hydroxide (NaOH). This causes the aluminum to selectively precipitate as aluminum hydroxide, which is removed as a solid (Velázquez-Martínez et al., 2019).

$$Al_2(SO_4)_{3(aq)} + 6\,NaOH_{(aq)} \rightarrow 2Al(OH)_{3(s)} + 3Na_2(SO_4)_{(aq)} \qquad \text{(Eq. 6-6)}$$

Step 3: Manganese (Mn) Precipitation

The remaining liquid is then treated with an oxidant, such as hydrogen peroxide (H_2O_2), in an alkaline environment. This oxidizes the soluble manganese (Mn^{2+}) to an insoluble manganese dioxide (Mn^{4+}) precipitate.

$$Mn(SO_4)_{(aq)} + H_2O_{2(aq)} + 2NaOH_{(aq)} \rightarrow MnO_{2(s)} + Na_2(SO_4)_{(aq)} + 2H_2O_{(l)} \text{ (Eq. 6-7)}$$

Step 4: Cobalt (Co) Separation (Solvent Extraction)

The solution, now containing Co, Ni, and Li, enters a solvent extraction circuit. It is mixed with an organic solvent containing the extractant Cyanex 272 (represented as HR). This extractant is highly selective for cobalt over nickel at a controlled pH. The cobalt ions are chemically exchanged into the organic phase (Eq. 6-8), leaving the nickel and lithium in the aqueous raffinate. The cobalt is then recovered from the loaded organic phase by "stripping" it with a fresh sulfuric acid solution (Eq. 6-9) (Mantuano et al., 2006).

Extraction:

$$Co^{2+}_{(aq)} + 2HR_{(org)} \leftrightarrow CoR_{2(org)} + 2H^{+}_{(aq)} \text{(Eq. 6-8)}$$

Stripping:

$$CoR_{2(org)} + 2H^{+}_{(aq)} \rightarrow Co^{2+}_{(aq)} + 2HR_{(org)} \text{(Eq. 6-9)}$$

Step 5: Nickel (Ni) Precipitation

The aqueous raffinate from the solvent extraction step, now free of cobalt, is treated with more sodium hydroxide to raise the pH. This precipitates the nickel as nickel hydroxide, which is recovered as a solid.

$$NiSO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Ni(OH)_{2(s)} + Na_2(SO_4)_{(aq)} \text{(Eq. 6-10)}$$

Step 6: Lithium (Li) Precipitation

The final solution, containing only lithium sulfate, is heated to approximately 60-80°C. Solid sodium carbonate (Na_2CO_3) is then added to precipitate the final product, solid lithium carbonate (Li_2CO_3), which has a lower solubility at higher temperatures (Zhang et al., 1998).

$$Li_2(SO_4)_{(aq)} + Na_2CO_{3(aq)} \xrightarrow{60-80℃} Li_2CO_{3(s)} + Na_2(SO_4)_{(aq)} \text{(Eq. 6-11)}$$

6.1 Rate Equations

The overall plant throughput is governed by the kinetics of two distinct processes: the biological regeneration of the lixiviant in a culture bioreactor and the chemical leaching of the black mass in a separate leaching circuit. The primary reaction governing the overall plant throughput is the biological regeneration of the Fe^{3+} lixiviant (Eq. 6-1). The rate of this reaction determines the required residence time and volume of the culture bioreactors, and consequently, the sizing of the associated air compression and aeration systems. As the overall process is a complex, multi-variable system, a simple kinetic model is insufficient for design. Therefore, the design of the bioreactor is based on the empirical, volumetric rate of reaction derived from the foundational process study. This empirical rate is calculated from the experimental data presented by Garcia et al. (2025). The data in Figure 3a of that study shows the Fe^{3+} concentration increasing from 0 g/L to 4.7 g/L over the initial 48-hour (2-day) bacterial growth phase. This corresponds to an average mass production rate of 0.0979 g/L/hr. Using the molar mass of iron (55.845 g/mol), this rate is converted to the final volumetric molar rate as follows:

$$\frac{0.0979 g/L*hr}{55.845 g/mol} * \frac{1000 mmol}{1 mol} = 1.75 mmol/L * hr \text{ (Eq. 6-12)}$$

This empirical rate of 1.75 mmol/L/h serves as the key kinetic parameter for designing the bioreactor system. The chemical leaching reactions (Eq. 6-2 through 6-4) are comparatively fast (on the order of minutes) and are therefore not the rate-determining step for the plant, provided a sufficient supply of Fe^{3+} lixiviant is maintained.

6.2 Empirical Rules

The plant operates under specific empirical conditions derived from the process study (Garcia et al., 2025). The process is designed to operate under ambient temperature (approx. 27-30°C) and atmospheric pressure. A highly acidic environment of pH 1.5 is required to keep the metals in solution and support the acidophilic bacteria. This pH is maintained by the controlled dosing of sulfuric acid. For the purposes of the plant's energy balance, the energy conversion is based on the chemical enthalpy of the metabolic reaction the bacteria catalyze. A. Ferrooxidans is a chemolithoautotrophic bacterium that gains its metabolic energy by oxidizing inorganic substrates. The standard enthalpy change (ΔH) for its primary metabolic reaction (Eq. 6-1) is approximately: $\Delta H \simeq -98$ kJ per mole of Fe^{2+} oxidized

This enthalpy is of similar magnitude to the Gibbs free energy change ($\Delta G^{\circ\prime}$), which is about –65 to –100 kJ/mol Fe^{2+} (Kelly & Wood, 2000; Rohwerder et al., 2003). This exothermic value represents the main heat source in the bioleaching circuit. This heat must be continuously removed by a dedicated heat exchange circuit to maintain the process at its optimal ambient temperature, as this relatively modest energy yield explains why the bacteria must oxidize large amounts of Fe^{2+} to sustain growth.

The Pregnant Leach Solution (PLS) from the bioleaching circuit contains a mixture of dissolved metal sulfates. The kinetics of their separation and recovery are not governed by the slow biological rates of the leaching step, but rather by the principles of chemical reaction engineering, mass transfer, and precipitation kinetics. Each step is designed to selectively isolate a target metal by controlling specific operating conditions.

6.2.1 Copper Cementation (Eq. 6-6)

The first recovery step is the cementation of copper, an electrochemical redox reaction between the sacrificial scrap iron (Fe^0) and the dissolved copper ions (Cu^{2+}). The rate of this heterogeneous reaction is typically controlled by the diffusion of Cu^{2+} ions from the bulk solution to the iron surface, or by a mixed control mechanism involving both diffusion and the charge transfer at the surface. As an empirical rule, the rate is directly proportional to the Cu^{2+}

concentration and is enhanced by increased agitation, higher temperature, and a larger surface area of the iron scrap. The process is often modelled using the shrinking-core model, where the fraction of copper recovered, X, is a function of time, t, and an apparent rate constant, k:

$$1 - (1 - X)^3 = kt \quad \text{(Eq. 6-13)}$$

6.2.2 Aluminum Precipitation (Eq. 6-7)

Following copper removal, aluminum is precipitated via hydrolysis as $Al(OH)_3$. The rate of this process is governed by nucleation and crystal growth kinetics. The most critical empirical rule is that the precipitation rate is strongly dependent on the pH, which must be tightly controlled within the 4.5–5.5 range to ensure selectivity. The nucleation rate is driven by the supersaturation ratio (S), and empirical rate laws for this type of precipitation are often written as:

$$R = k * (S - 1)^n \quad \text{(Eq. 6-14)}$$

Where R is the precipitation rate, S is the supersaturation ratio, and n is an empirical exponent, typically between 1 and 2.

6.2.3 Manganese Precipitation (Eq. 6-8)

Manganese is removed through the chemical oxidation of soluble Mn^{2+} to solid MnO_2 using hydrogen peroxide (H_2O_2) in an alkaline solution. The rate is controlled by the chemical kinetics of oxidation. As an empirical rule, the rate is proportional to the concentrations of both [Mn^{2+}] and [H_2O_2], and it is strongly dependent on pH; the reaction proceeds faster at higher pH levels, which helps stabilize the MnO_2 nuclei. When H_2O_2 is supplied in stoichiometric excess, the kinetics can often be modelled as pseudo-first order with respect to the Mn^{2+} concentration.

6.2.4 Cobalt Solvent Extraction (Eq. 6-9, 6-10)

Cobalt is selectively separated from nickel and lithium using a solvent extraction (SX) circuit. This involves a reversible ion exchange between the aqueous Co^{2+} ions and the organic extractant, Cyanex 272. The rate of this process is complex, as it is controlled by a combination of interfacial mass transfer (the area, A, where the oil and water meet) and the chemical complexation reaction at that interface. The extraction kinetics are often modelled as:

$$R = k * A * (C_{aq} - C_{eq}) \quad \text{(Eq. 6-15)}$$

Where k is the mass transfer coefficient, C_{aq} is the aqueous cobalt concentration, and C_{eq} is the concentration at equilibrium. The dominant empirical rule for this process is pH control, as the selectivity of Cyanex 272 for cobalt over nickel is highly pH-dependent. The subsequent stripping reaction (Eq. 6-10) is typically rapid and considered to be equilibrium-controlled.

6.2.5 Nickel Precipitation (Eq. 6-11)

The aqueous raffinate from the cobalt SX circuit, now containing nickel and lithium, is treated to precipitate nickel hydroxide. Similar to the aluminum step, the rate is controlled by supersaturation and nucleation kinetics. The key empirical rule is that the rate increases dramatically as the pH is raised above 8. The process follows a similar supersaturation-based rate law, and the final particle size distribution of the $Ni(OH)_2$ product is heavily influenced by the mixing intensity and reagent addition rate within the precipitator.

6.2.6 Lithium Precipitation (Eq. 6-12)

The final recovery step involves precipitating lithium carbonate (Li_2CO_3) from the nickel-free solution by adding sodium carbonate (Na_2CO_3). The rate is controlled by nucleation and crystal growth kinetics. The most important empirical rule for this step is temperature, as Li_2CO_3 exhibits inverse solubility (its solubility decreases as temperature increases). The process is therefore operated at an elevated temperature (60–80°C) to maximize the supersaturation and drive a higher product yield. The rate is proportional to this supersaturation, and the process is industrially modelled using population balance equations to predict the final crystal size distribution

Chapter 7. The Process

7.1 Detailed Process Flow Diagram

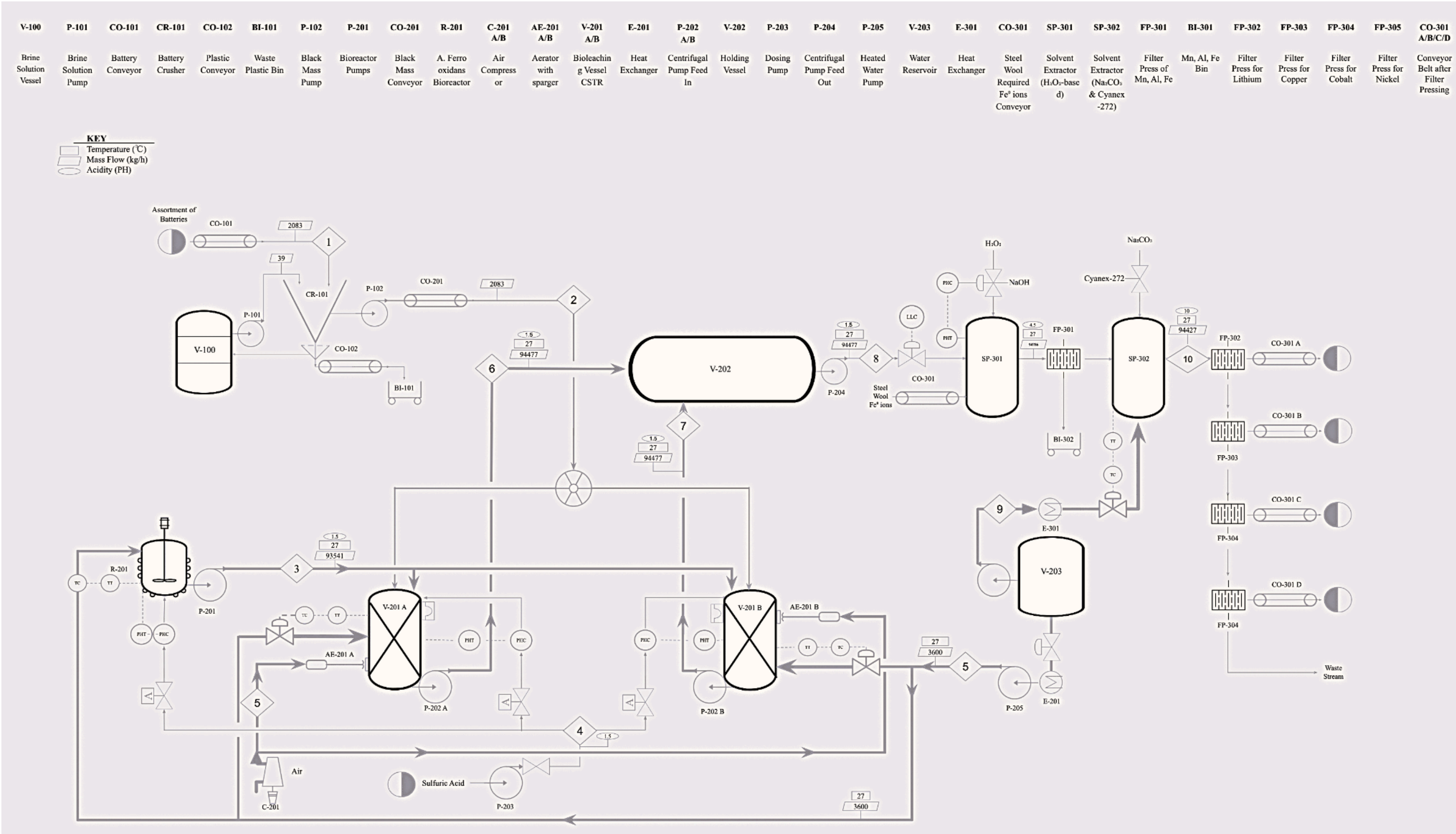

FIGURE 7.1 DETAILED PFD

Table 7.1.1 below shows all the equipment used in the process, alongside their respective codes.

Table 7.1.1 PFD Equipment and respective codes

Code	Equipment Name	Code	Equipment Name	Code	Equipment Name
V-100	Brine Solution Vessel	**AE-201 A/B**	Aerator with sparger	**SP-301**	Solvent Extractor (H_2O_2-based)
P-101	Brine Solution Pump	**V-201 A/B**	Bioleaching Vessel (CSTR)	**SP-302**	Solvent Extractor (Na_2CO_3 & Cyanex-272)
CO-101	Battery Conveyor	**E-201**	Heat Exchanger	**FP-301**	Filter Press of Mn, Al, Fe
CR-101	Battery Crusher	**P-202 A/B**	Centrifugal Pump Feed In	**BI-301**	Mn, Al, Fe Bin
CO-102	Plastic Conveyor	**V-202**	Holding Vessel	**FP-302**	Filter Press for Lithium
BI-101	Waste Plastic Bin	**P-203**	Dosing Pump	**FP-303**	Filter Press for Copper
P-102	Black Mass Pump	**P-204**	Centrifugal Pump Feed Out	**FP-304**	Filter Press for Cobalt
P-201	Bioreactor Pumps	**P-205**	Heated Water Pump	**FP-305**	Filter Press for Nickel
CO-201	Black Mass Conveyor	**V-203**	Water Reservoir	**CO-301 A/B/C/D**	Conveyor Belt after Filter Pressing
R-201	A.Ferrooxidans Bioreactor	**E-301**	Heat Exchanger		
C-201 A/B	Air Compressor	**CO-301**	Conveyor		

Table 7.1.2 indicates all the streams and their respective properties.

Table 7.1.2 PFD Streams and respective properties

Stream	Temperature(℃)	Mass Flow (kg/hr)	Acidity (pH)
1	Ambient	2083	N/A
2	Ambient	2083	N/A

3	27	93451	1.5
4	Ambient	pH controlled	1.5
5	27	3600	N/A
6	27	94477	1.5
7	27	94477	1.5
8	27	94477	1.5/4.5
9	27	Temperature Controlled	6.5-8.5*
10	27	94477	10

- Cited from *GB 5749-2022 Standards for Drinking Water Quality*. Beijing.

7.2 Plant Safety

A robust safety and loss prevention strategy is a foundational component of this plant design, particularly given the hazardous nature of the feedstock. The design philosophy prioritizes Inherently Safer Design (ISD), aiming to first eliminate or reduce hazards at their source, rather than just controlling them with add-on safety systems.

The most significant safety strategy was the selection of bioleaching itself. By choosing a low-energy, ambient-temperature process, the design has *inherently eliminated* the single greatest hazard of competing designs: the 1,400°C+ pyrometallurgical furnace. This ISD choice eliminates the risks associated with high-temperature operations, molten metal handling, and the generation of toxic off-gases, such as dioxins and furans (Roy et al., 2021). Furthermore, the process employs the minimal pre-treatment (5 mm particle) method (Garcia et al., 2025), which eliminates the need for fine grinding and thereby mitigates the risk of creating a combustible dust cloud.

Despite the ISD benefits, significant hazards remain, primarily concentrated in three plant areas identified on the PFD. The Pre-Treatment (Area 100) presents the highest risk. The Battery Crusher (CR-101) and associated conveyors (CO-101, CO-201) handle high-energy cells that can undergo thermal runaway, leading to fire and explosion upon puncture (Kim & Park, 2025). This stage also carries the risk of releasing highly toxic and corrosive Hydrogen Fluoride (HF) gas as the electrolyte is exposed. The second hazardous area is the Leaching Zone (Area 200), which includes the Bioleach Vessels (V-201 A/B). This area involves handling large volumes of corrosive Sulfuric Acid (H_2SO_4) via the Dosing Pump (P-203), which creates a risk of chemical burns from leaks. The third area is Recovery (Area 300), where the Solvent Extractors (SP-301, SP-302) handle both corrosive bases (NaOH) and flammable organic solvents (Cyanex 272), introducing a fire and VOC emission risk.

To manage these identified hazards, a multi-layered loss prevention strategy is implemented. A key passive safety system is the brine discharge bath in the Brine Solution Tank (V-100) (fed by P-101), which de-energizes all incoming batteries before they reach the crusher (CR-101). All chemical and solvent processing equipment (V-201 A/B, SP-301, SP-302) will be located within reinforced concrete spill containment bunds sized to hold 110% of the largest tank's volume. The CR-101 enclosure will be kept under negative pressure and vented to an acid gas scrubber. Active safety systems include thermal cameras and an automated nitrogen flooding system for the CR-101 vault, as well as instrumentation interlocks on the acid Dosing Pump (P-203) to prevent overdosing. Finally, procedural controls will include mandatory chemical-resistant PPE, a formal HAZOP (Hazard and Operability Study), and strict safe-work permitting.

7.2.1 Safety Equipment

The plant will be equipped with extensive safety hardware to protect personnel and the environment. This includes emergency showers and eyewash stations located within 10 seconds of travel from all chemical handling areas (including P-203, V-201 A/B, SP-301, and SP-302). A dedicated acid gas scrubber will treat all air exhausted from the CR-101 shredder enclosure, supplemented by an automated nitrogen flooding system for fire suppression in that same vault. To protect against over-pressure, rupture disks will be installed on the filter presses (FP-301 through FP-305), and conservation vents will be placed on all bulk storage tanks. The process areas will be monitored by a network of fixed-point toxic gas detectors (HF and VOC), and mobile spill kits with appropriate neutralizers will be staged in all process areas.

7.3 Special Provisions

In addition to the primary safety systems outlined in Section 7.2, the plant design incorporates several special provisions to mitigate specific, localized hazards related to noise, fugitive emissions, and fire risk. These provisions are concentrated in the pre-treatment and solvent extraction areas of the plant.

A dedicated, special enclosure will be constructed to house the Battery Crusher (CR-101), the Battery Conveyor (CO-101), and the Black Mass Conveyor (CO-201). This enclosure serves multiple critical functions. First, it serves as an acoustic barrier, constructed from reinforced concrete and sound-dampening panels to attenuate the high noise levels (>100 dB) generated by the crushing operation. Second, it acts as a primary containment and ventilation zone. The enclosure will be maintained under negative pressure, with all exhausted air routed to a dedicated acid gas scrubber to capture any fugitive dust or toxic Hydrogen Fluoride (HF) gas released from ruptured cells.

Due to the risk of flammable electrolyte vapours in the crushing enclosure and the presence of organic solvents in the recovery area, spark-proof electrical wiring and explosion-proof (XP) rated equipment will be mandatory in these designated zones. This includes all motors for the crusher (CR-101), conveyors (CO-101, CO-201, CO-301), and all pumps (P-102, P-202, etc.), lighting, and instrumentation located within the classified boundaries of the Pre-Treatment (Zone 100) and Recovery (Zone 300) areas.

Noise control is also a provision for major rotating equipment. The Air Compressors (C-101 A/B), which supply the Aerators (AE-201 A/B), are significant noise sources. They will be housed in their own dedicated, sound-insulated building to protect personnel in adjacent areas of the plant.

Finally, provisions are made for odour and bio-aerosol control. The off-gas from the aerated Bioleach Vessels (V-101 A/B) and the Culture CSTR (R-101) will not be vented directly to the atmosphere. This exhaust stream will be collected in a header and routed through a bio-filter to remove any potential odours (e.g., hydrogen sulfide) or entrained bacterial aerosols before release. The Solvent Extractor units (SP-301, SP-302) will be covered and vented to a vapour recovery system to minimize fugitive VOC emissions from the organic solvents.

7.4 Mass Balance

The plant is required to recycle 15,000 tons/year of batteries. It is assumed that the plant will operate 24 hours a day, 7 days a week, for 300 days a year. This would mean that the plant is required to process batteries at a rate of 2083.33 kg/h.

7.4.1 Mass Balance of The Crusher

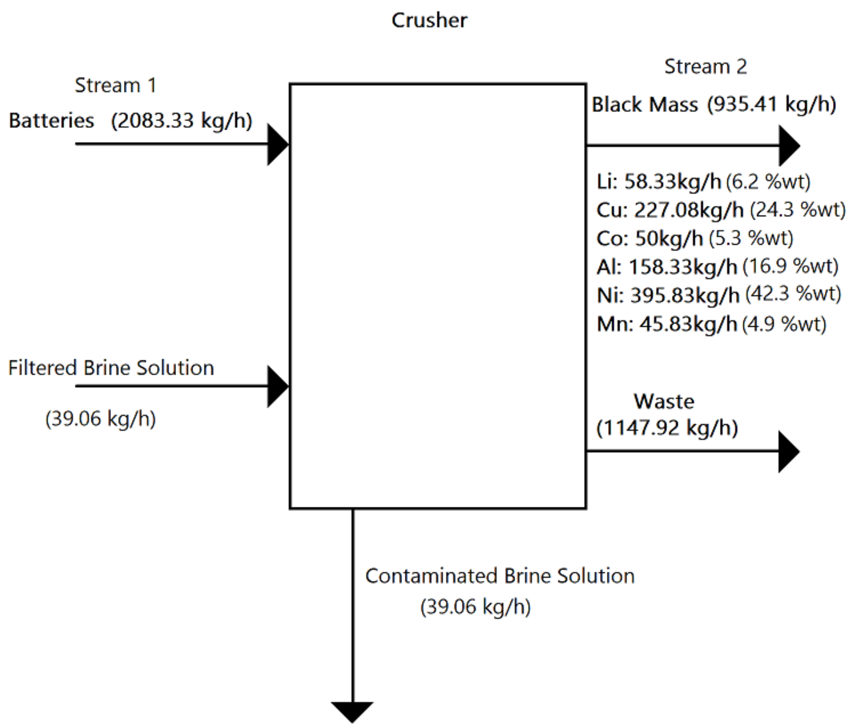

FIGURE 7.4.1: CRUSHER MASS BALANCE

Adapted from Garcia et al. (2025), we know the percent recovery of metal after crushing, which determines the composition of black mass to be bioleached by A. Ferrooxidans. From Table 7.4.1, we can see that the total mass flow to process batteries is at a rate of 2083.33 kg/h.

We require a brine solution of 39kg/h to discharge all batteries while crushing and preventing any combustion, as the batteries are crushed in an inert environment. As such, we obtain;

TABLE 7.4.1 CRUSHER OUTPUT

output	Recovery %	kg/h
Waste	55.10%	1147.92
Li	2.80%	58.33
Cu	10.90%	227.08
Co	2.40%	50
Al	7.60%	158.33
Ni	19%	395.83
Mn	2.20%	45.83

We can observe that a majority of the output stream presents as the waste stream. This is due to the crushing of unprocessed batteries, which contain all other materials enclosed within them. The waste stream includes all plastics, polymers, and "waste" present after crushing many kinds of batteries. After crushing, we can determine an estimated product stream of metals within the output stream of the crusher. These properties are used as the basis to determine the metals leached in the biohydrometallurgical process of bioleaching using A. Ferrooxidans in the CSTR Figure 7.4.2.

TABLE 7.4.2 BRINE SOLUTION

Variable	Amount	Units
Liquid holdup	3	3:1 brine to bat ratio
Brine Flow	6250	kg brine solution
brine density	1050	kg/m^3
wt%	10	% NaCL
Temperature	23	C
purge %	15.00%	purge%/day
total flow	39.0625	kg/h

A brine solution is used with the input of the crusher to ensure safe, complete discharge of the batteries before and after crushing. The brine solution is determined by a 3 to 1 ratio of mass to mass flow. To prevent insufficient discharge of batteries, our solution is to oversize, which accounts for inconsistent battery chemistry. Thus, at a given temperature and purge percent, we need a total flow of about 39kg/h of brine to ensure complete discharge and submersion of batteries.

7.4.2 Mass Balance of The CSTR

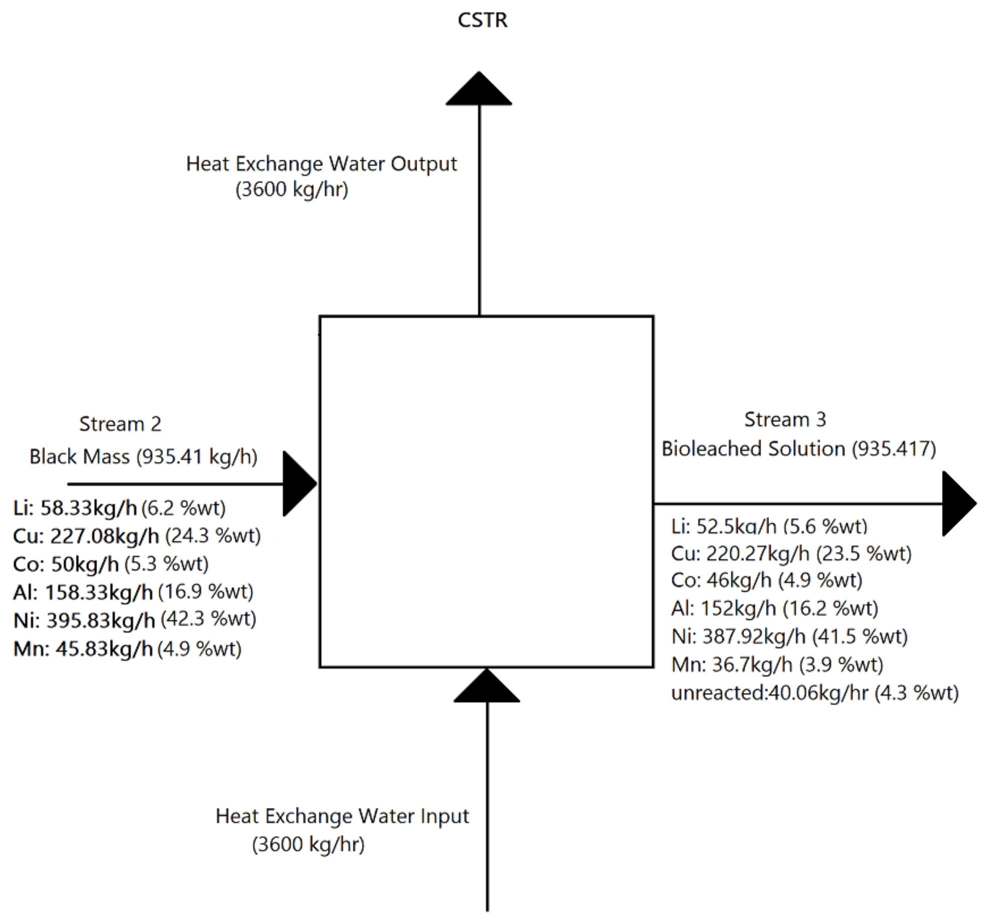

FIGURE 7.4.2: CSTR MASS BALANCE

The Bioleaching Continuously Stirred-Tank Reactor (CSTR) is the primary unit operation for the chemical dissolution of metals. This unit receives the solid Black Mass from the pre-treatment circuit at a continuous rate of 935.41 kg/h. Inside the reactor, the black mass is mixed with a biogenic lixiviant (leaching agent) consisting of bio-oxidized ferric iron (Fe^{3+}) generated by Acidithiobacillus ferrooxidans. This process operates at a steady state to dissolve the target metals from the solid phase into an aqueous liquid phase, known as the Pregnant Leach Solution (PLS).

The CSTR unit separates the feed into two distinct output streams. The primary product stream is the Bioleached Solution, a liquid leachate containing 895.35 kg/h of dissolved metal sulfates, which is sent forward to the metals recovery circuit. The second output is the Solid Waste, a 40.06 kg/h stream of unreacted solids, primarily composed of graphite, binder, and any un-leached metal oxides. A separate utility stream of heat exchange water, with a flow rate of 3,600 kg/h, is used to maintain the reactor at its optimal ambient temperature but does not mix with the process streams and thus does not factor into the process mass balance.

Table. 7.4.3 CSTR Output

Component	Recovery %	Flow Rate
Li	90%	52.50
Cu	97%	220.27
Co	92%	46.00
Al	96%	152.00
Ni	98%	387.92
Mn	80%	36.67

The distribution of mass and the recovery efficiencies for each metal are detailed in the component mass balance. The calculations are based on experimental data from the foundational bioleaching study by Garcia et al. (2025). The process achieves a 90% recovery of Lithium, resulting in 52.5 kg/h in the leachate. For Copper, a 97% recovery is achieved, sending 220.27 kg/h to the recovery circuit. Cobalt is recovered at 92%, yielding 46.0 kg/h in the leachate, while Aluminum is recovered at 96% (152.0 kg/h). The highest recovery is seen for Nickel at 98%, corresponding to a liquid-phase flow of 387.92 kg/h. Manganese is recovered at an 80% rate, resulting in 36.67 kg/h of dissolved metal. The remaining 40.06 kg/h of unreacted material is discharged as the solid waste stream.

These streams are able to recover these rates due to the input of the culture medium of A. Ferrooxidans. Allowing for efficient bioleaching to occur within the CSTR at room temperature and standard conditions.

TABLE. 7.4.4 BIOREACTOR OUTPUT OF CULTURE MEDIUM

Culture Medium		
amount needed	93541.667	kg/h
composition		
$(NH_4)_2SO_4$	3.000	g/L
K_2HPO_4	0.500	g/L
$MgSO_4 \cdot 7H_2O$	0.500	g/L
KCl	0.100	g/L
$Ca(NO_3)_2 \cdot 4H_2O$	0.014	g/L
Fe	8.900	g/L
composition (kg/h) (density is 1Kg/L)		
$(NH_4)_2SO_4$	280.625	kg/h
K_2HPO_4	46.771	kg/h
$MgSO_4 \cdot 7H_2O$	46.771	kg/h
KCl	9.354	kg/h
$Ca(NO_3)_2 \cdot 4H_2O$	1.310	kg/h
Fe	832.521	kg/h
composition of mass flow rate out of the reactor		

The "Culture Medium" stream, which enters the bioleaching CSTR at a rate of 93,541.67 kg/h, is the active leaching agent that contains the A. Ferrooxidans bacteria and all the nutrients required for their metabolic process. This solution is continuously generated in the upstream bioreactors and fed to the CSTR to react with the Black Mass. As shown in Table 7.4.4, the medium's composition is based on the 6K mineral medium described in the foundational process study (Garcia et al., 2025).

The most critical component is the 832.52 kg/h of Iron (Fe). This is fed as ferrous sulfate (Fe^{2+}), which acts as the "food" or energy source for the A. Ferrooxidans bacteria. As detailed in

Chapter 6 (Eq. 6-1), the bacteria oxidize this Fe^{2+} into ferric iron (Fe^{3+}). This Fe^{3+} is the powerful lixiviant that chemically attacks the Black Mass, dissolving the valuable metals.

The other components, such as ammonium sulfate and potassium phosphate, are essential trace nutrients that support the health and reproduction of the bacterial colony. Therefore, the CSTR's primary function is to mix this active, iron-rich culture medium with the metal-rich Black Mass to facilitate the rapid leaching reactions.

7.4.3 Separator 1

This control volume, designated as the first separation stage, follows the bioleaching CSTR. Its primary purpose is to purify the main leachate stream by selectively removing specific metal impurities (aluminum, manganese) and recovering the first valuable product (copper), thereby preparing a purified solution for the subsequent cobalt and nickel recovery circuits.

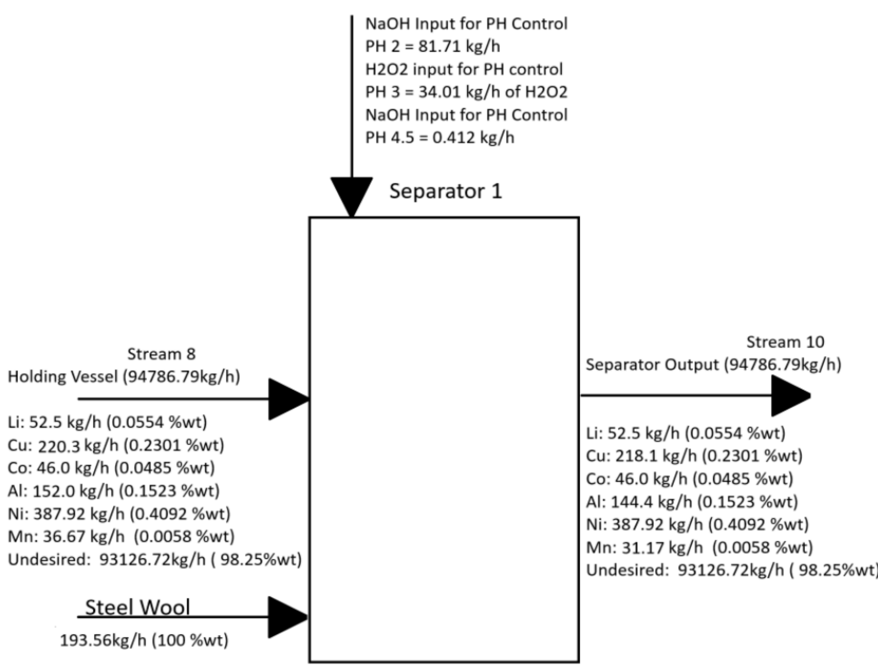

FIGURE: 7.4.3 BLOCK DIAGRAM OF SEPARATOR 1

TABLE. 7.4.5 MASS FLOW RATES AFTER SEPARATOR 1 PROCESSING

Component	Amount	Units
Li	52.50	kg/h
Cu	220.27	kg/h
Co	46.00	kg/h
Al	152.00	kg/h
Ni	387.92	kg/h
Mn	36.67	kg/h
Unreacted	40.06	kg/h
Culture Medium	93541.67	kg/h
Fe^0	193.56	kg
NaOH (PH 2)	81.71	kg
H_2O_2 (PH 3)	34.01	kg
NaOH (PH 4.5)	0.4117	kg

This unit receives two process streams: the Bioleached Solution, which contains 895.35 kg/h of dissolved metals in 93,541.67 kg/h of culture medium, and the Solid Waste from the CSTR, which consists of 40.06 kg/h of unreacted solids (e.g., graphite).

To achieve the desired separations, three reagent streams are introduced. First, 193.56 kg/h of Iron powder is added to perform cementation of copper, as described in Chapter 6 (Eq. 6-6). Second, 34.01 kg/h of Hydrogen Peroxide is dosed as an oxidant to facilitate the precipitation of manganese (Eq. 6-8). Finally, 82.12 kg/h of Sodium Hydroxide is added as a base to control the pH and drive the precipitation of aluminum (Eq. 6-7) and manganese.

This process yields two distinct output streams. The first is the Solid Precipitate Stream, which is sent to a filter press for dewatering. This solid stream has a total mass flow of 1,300.20 kg/h. It contains the unreacted solids from the CSTR (40.06 kg/h), the precipitated iron (832.52 kg/h), and the target recovered metals. The separation is highly effective, capturing 99% of the incoming copper (218.07 kg/h), 95% of the aluminum (144.4 kg/h), and 85% of the manganese (31.17 kg/h).

The second output is the Purified Leachate, which proceeds to the next recovery stage at a total flow rate of 94,390.26 kg/h. This stream's critical value lies in what it retains: 100% of the Lithium (52.5 kg/h), 100% of the Cobalt (46.0 kg/h), and 100% of the Nickel (387.92 kg/h). This purified solution now also contains the unrecovered fractions of the upstream metals (2.20 kg/h Cu, 7.6 kg/h Al, and 5.5 kg/h Mn) as well as the dissolved byproducts from the added reagents, chiefly 191.63 kg/h of dissolved iron and 66.75 kg/h of sodium ions.

7.4.4 Separator 2

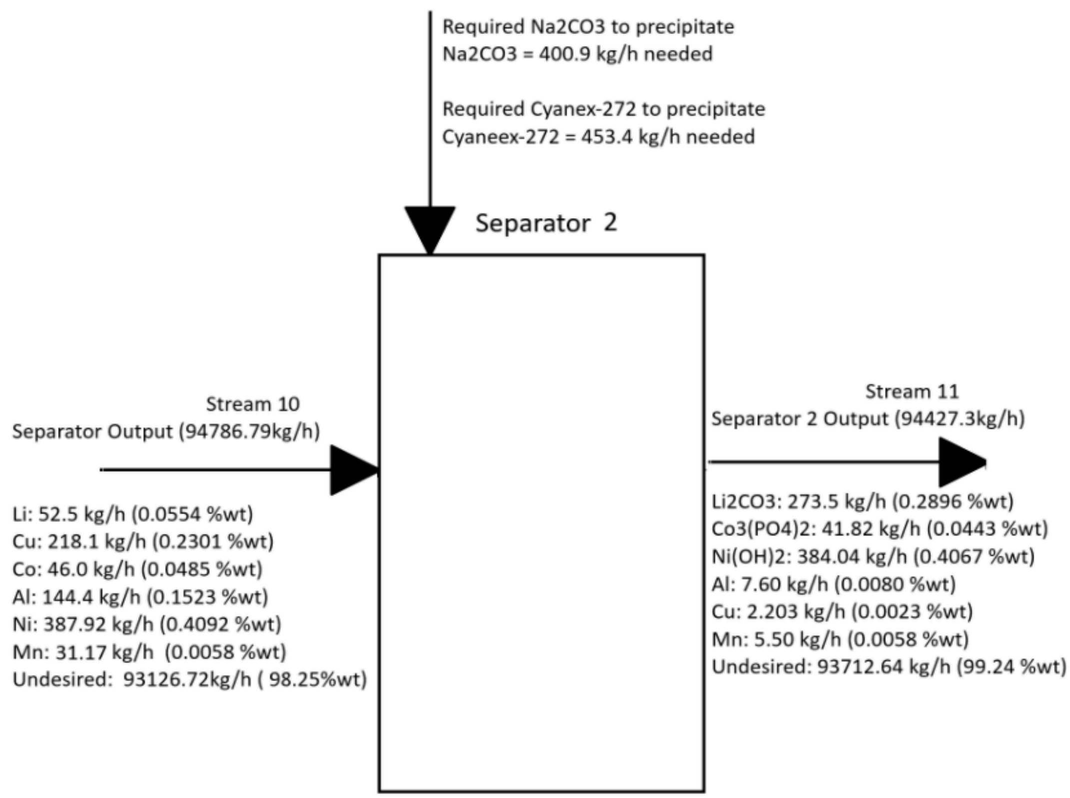

FIGURE: 7.4.4 BLOCK DIAGRAM OF SEPARATOR 2

This control volume, designated as the second separation stage, receives the purified leachate stream from the impurity precipitation circuit. The total mass flow into this unit is 94,427.28 kg/h. This stream contains the valuable dissolved metals, primarily 52.5 kg/h of Lithium, 46.0 kg/h of Cobalt, and 387.92 kg/h of Nickel, which are the targets for recovery in this block. This stage is a complex, multi-step recovery circuit involving the addition of several key reagents: 453.37 kg/h of Cyanex-272 (for solvent extraction), 0.12 kg/h of Sodium Hydroxide (NaOH), and 400.90 kg/h of Sodium Carbonate (Na_2CO_3) (for precipitation).

The process executes three sequential separations to recover the final metal products. First, cobalt is selectively recovered using the Cyanex-272 solvent. This extraction is highly efficient, recovering 90.91% of the inlet cobalt, which corresponds to a product stream of 41.82 kg/h of Cobalt Phosphinate. The remaining 4.18 kg/h of cobalt is not recovered and passes to the subsequent step.

Next, the cobalt-free raffinate is treated to recover nickel. This precipitation step demonstrates a 99% recovery efficiency, separating 384.04 kg/h of Nickel Hydroxide ($Ni(OH)_2$) as a solid product. The remaining 1% of nickel (3.88 kg/h) is lost to the final effluent.

TABLE. 7.4.6 MASS FLOW RATES AFTER SEPARATOR 2 PROCESSING

Component	Amount (kg/h)
Fe^{2+}	191.63
Fe^{0}	1.94
Unfiltered waste	12.26
Li	52.50
Cu	2.20
Co	46.00
Al (in sol.)	7.60
Ni	387.92
Mn (in sol.)	5.50

Finally, the remaining solution enters the lithium recovery circuit. This stage achieves a 98% recovery of the inlet lithium, precipitating 273.49 kg/h of Lithium Carbonate (Li_2CO_3) as the final product. The unrecovered 1.05 kg/h of lithium is lost in the final wastewater stream. Other dissolved components from the previous stage, such as 2.20 kg/h of Copper and 7.6 kg/h of Aluminum, are not targeted by these reagents and pass through the system, exiting with the final aqueous raffinate.

After the Separators, the solution is fed onto a series of filter presses which are responsible in extracting the metals in their final solid form(cakes). The filter presses are arranged in a pattern such that each unit is responsible in separating a different component from the other.

From Filter Press 1 and 2 we obtain all our desired products as dry cake mass as follows in table 7.4.7.

Table. 7.4.7 Filter Press 1 & 2 obtained metals and recovery Percentage

Component	Recovery %	Amount (kg/h)
Al (out of sol.)	99%	142.96
Mn (out of sol.)	99%	30.86
Cu (out of sol.)	99%	215.89
Li_2CO_3	99%	270.75
$Co_3(PO_4)_2$	99%	41.40
$Ni(OH)_2$	99%	380.20

As a result of our plant design process we obtain solid Al, Mn, Cu, out of filter press 1 and 2 gives Li_2CO_3, $Co_3(PO_4)_2$, and $Ni(OH)_2$.

7.4.5 Overall Mass Balance

Using all the information calculated from the mass balance of the independent parts, the table below outlines the control volume of the entire system

TABLE. 7.4.8 OVERALL MASS BALANCE

Overall Mass Balance in	
in	kg/h
battery	2083.3
brine solution	39.062
Fe^0	193.56
NaOH	116.25
H2O2	34.012
Cyanex-272	453.37
Na2CO3	400.90

Overall Mass Balance out	
out	kg/h
Fe2+	191.63
Fe^0	1.9356
unfiltered waste	12.262
Li2CO3	270.75
Li2CO3 (in sol.)	2.7349
unreacted Na2CO3	8.0180
Li (in sol.)	1.05
Cu	2.2027
Cobalt phosphinate	41.4
Co. phosph. (in sol.)	0.41818
Co (in sol.)	4.1818
Al	7.6
Ni(OH)2	380.20
Ni(OH)2 (in sol.)	3.8404
Ni (in sol.)	3.8792
Mn	5.5
Unreacted	40.062
Culture Medium	92709
Na^+	237.66
unreacted Cyanex-272	41.215
H^+	412.15
Water created	49.438

7.5 Energy Balance

All major calculations are sampled in the Appendix under energy balance. Here are the presented control volumes around overall equipment and their energy requirements. Of each month's energy consumption based on average temperature conditions in Shenzhen China.

7.5.1.1 Energy Balance on Crusher and Air compressor

The crusher is assumed to be able to handle a power usage of 298.28kW of energy to crush our desired input of 2083.33 kg/h of unsorted secondary batteries. The air compressor is

7.5.1.2 Energy Balance on Bioleaching CSTR

A. Ferrooxidans do not have a single heat of reaction value like a chemical compound. Instead, its metabolism is based on the oxidation of ferrous iron ($Fe^{2+} \rightarrow Fe^{3+}$) or reduced sulfur compounds, and the enthalpy change of these redox reactions provides the energy the organism uses. The standard enthalpy change for the oxidation of Fe^{2+} to Fe^{3+} under acidic conditions is approximately –98 kJ per mole of Fe^{2+} oxidized (at 25 °C, 1 M activity), which represents the biological energy yield available to A. Ferrooxidans (Kelly & Wood, 2000; Rohwerder et al., 2003). A. Ferrooxidans is a chemolithoautotrophic bacterium: it gains energy by oxidizing inorganic substrates, primarily ferrous iron (Fe^{2+}) and reduced sulfur species.

The key metabolic reaction is:

$$Fe^{2+} \rightarrow Fe^{3+} + e^{-}$$

Coupled with oxygen reduction:

$$\frac{1}{4}O_2 + H^{+} + e^{-} \rightarrow \frac{1}{2}H_2O$$

When combined, the overall reaction is:

$$Fe^{2+} + \frac{1}{4}O_2 + H^{+} \rightarrow Fe^{3+} + \frac{1}{2}H_2O$$

The Gibbs free energy change ($\Delta G^{\circ\prime}$) for this reaction is about –65 to –100 kJ/mol Fe^{2+}, depending on pH and redox potential (Kelly & Wood, 2000). The enthalpy (ΔH) is of similar magnitude, around –98 kJ/mol Fe^{2+} under standard conditions. This relatively modest energy yield explains why A. Ferrooxidans must oxidize large amounts of Fe^{2+} to sustain growth.

This reaction is conducted at ambient temperature (~27°C) and pressure, with aeration to provide the necessary oxygen. The kinetics of this reaction govern the plant's overall throughput, as a continuous supply of Fe^{3+} is required.

Based on an analysis of the experimental data published by Garcia et al. (2025), the average volumetric reaction rate during the most active phase of bacterial growth was calculated. This empirical rate is derived directly from the experimental data in the primary process study (Garcia et al., 2025). The data in Figure 3a of that study shows the Fe^{3+} concentration increasing from 0 g/L to 4.7 g/L over the initial 48-hour (2-day) bacterial growth phase. This corresponds to an average mass production rate of 0.0979 g/L/hr. Using the molar mass of iron (55.845 g/mol), this rate is converted to the final volumetric molar rate as follows:

$$\frac{0.09979\ g/L.hr}{55.845\ g/mol} \times \frac{1000\ mmol}{1\ mol} = 1.75\ mmol/L/hr$$

also assumed to be able to run with 200kW of energy used to aerate and the CSTR's optimal conditions. Table 7.5.1.2 shows the temperature of water needed for each month in the jacket.

TABLE: 7.5.1.2 Energy Balance on Bioleaching CSTR Results

	Bioleaching vessel (A) , V-101 A [A. Ferrooxidans]							
months	Feed (kW)	Air (kW)	evap(cooling)	Agitator (kW)	rxn/meta (kW)	Wall loss (kW)	Q jacket (kW)	T needed (°C)
1	-1.1172	-0.86	0.015	20	2.381	-44.40	23.982	36.346
2	-0.88445	-0.68	0.290	20	2.381	-35.15	14.045	32.473
3	-0.6517	-0.50	0.292	20	2.381	-25.90	4.380	28.707
4	-0.3724	-0.29	0.295	20	2.381	-14.80	-7.217	24.187
5	0	0.00	0.299	20	2.381	0.00	-22.681	18.161
6	0.0931	0.07	0.300	20	2.381	3.70	-26.547	16.654
7	0.1862	0.14	0.301	20	2.381	7.40	-30.412	15.148

8	0.13965	0.11	0.301	20	2.381	5.55	-28.480	15.901
9	0.093	0.07	0.300	20	2.381	3.70	-26.547	16.657
10	-0.186	-0.14	0.297	20	2.381	-7.40	-14.949	21.174
11	-0.558	-0.43	0.293	20	2.381	-22.20	0.514	27.200
12	-0.977	-0.75	0.289	20	2.381	-38.85	17.911	33.980

7.5.1.3 Energy Balance on Separator 2

In order for optimal Li recovery, the reaction that uses Na_2CO_3 to precipitate the Li out as Li_2CO_3 is best conducted at 60°C. The feed is assumed to be 27°C from the CSTR, and there are slight energy losses from the walls. This means that the required water temperature to keep the temperature at 60°C is shown in the table below.

Table: 7.5.1.3 Energy Balance on Separator 2 results

Control volume	**Separator 2**			
months	feed (kW)	Wall loss (kW)	Q jacket (kW)	T needed (°C)
1	-3.5489	-166.5	170.049	106.733
2	-3.5489	-157.25	160.799	104.191
3	-3.5489	-148	151.549	101.650
4	-3.5489	-136.9	140.449	98.599
5	-3.5489	-122.1	125.649	94.531
6	-3.5489	-118.4	121.949	93.514
7	-3.5489	-114.7	118.249	92.497
8	-3.5489	-116.55	120.099	93.006
9	-3.5489	-118.4	121.949	93.514
10	-3.5489	-129.5	133.049	96.565
11	-3.5489	-144.3	147.849	100.632
12	-3.5489	-160.95	164.4999	105.208

7.5.1.4 Energy Balance on Compressor

The calculation of the Energy balance was done by scaling up the compressor used in Garcia et al. (2025). The sized up value came out to be 200kW.

7.5.1.5 Energy Balance on Pump

The energy of the pump is dependent on the volumetric flow rate and the differential head pressure. Because it is required to size to find the differential head pressure, it is assumed to equal 1m.

TABLE: 7.5.1.5 ENERGY BALANCE ON PUMP RESULTS

control volume	**Pump**							
	P-101 (kW)	P-102 (kW)	P-201 (kW)	P-202 (kW)	P-203 (kW)	P-204 (kW)	P-205 (kW)	P-206 (kW)
energy needed	0.1064	2.5490	509.80	128.72	0.2725	257.45	9.81	9.81

7.5.1.6 Energy Balance on Conveyor

The energy balance on the conveyor systems quantifies the total power requirement for each control volume (conveyor unit).

This required total energy, expressed in kilowatts (kW), is the sum of several distinct power components: the power needed for lifting the material (P_{lift}), the power lost due to rolling resistance of the material on the belt ($P_{roll,mat}$), the power lost due to belt rolling resistance on the idlers ($P_{roll,belt}$), the power transmitted through the shaft (P_{shaft}), and the power supplied by the motor (P_{motor}).

The results, as shown in the table, indicate the specific power consumption for each conveyor unit to perform its transport task, allowing for the calculation of overall energy usage and efficiency within the process.

Table: 7.5.1.6 Energy Balance on Conveyor Results

Control volume	Conveyor					
conveyor	P lift (W)	P roll,mat (W)	P roll,belt (W)	P shaft (W)	P motor (W)	total energy (kW)
CO-101	28.385	0.7096	2.1459	31.241	34.712	0.03471
CO-102	15.640	0.3910	2.1459	18.177	20.197	0.02020
CO-201	12.745	0.3186	2.1459	15.210	16.899	0.01690
CO-301	2.6373	0.0659	2.1459	4.8492	5.3879	0.00538
CO-301 A,B,C,D	5.1801	0.1295	2.1459	7.4556	8.2840	0.00828

7.5.1.7 Energy Balance on Heat Exchanger

The two heat exchangers are used to heat the water that is used as a cooling jacket in the CSTR and the separator. Using earlier calculations, it is known what temperature of water is needed to stay at optimum temperature in vessels. This means that the energy calculation varies per month because the water is assumed to be the same temperature as the environment.

Table: 7.5.1.7 Energy Balance on Heat Exchanger Results

Control volume	Heat Exchanger	
Months	E-201 (kW)	E-301 (kW)
1	89.354	383.99
2	62.679	362.89
3	36.448	341.78
4	4.970	316.45
5	-37.001	282.68
6	-47.493	274.24
7	-57.986	265.80
8	-52.740	270.02
9	-47.493	274.24

10	-16.015	299.57
11	25.958	333.34
12	73.172	371.33

7.5.1 Energy Efficiency

To minimize energy consumption in the plant, several passive and process-integrated strategies were implemented. These choices reduce reliance on mechanical systems, improve thermal efficiency, and enhance operational resilience. Below are the key steps taken, with emphasis on gravity-driven flow and a suggestion for an additional efficiency measure:

To reduce the overall energy consumption of the plant, several integrated design strategies were implemented that prioritize passive energy recovery, natural circulation, and gravity-assisted flow. These measures not only improve thermal and hydraulic efficiency but also enhance operational reliability and reduce dependence on mechanical systems.

One of the most significant steps taken was the use of gravity-driven flow through vertical equipment placement. In this configuration, process vessels such as leach reactors, clarifiers, and precipitation tanks are arranged in a tiered layout, allowing liquids to move from one stage to the next without the need for pumps. To further optimize this setup, two vessels are used in parallel at each stage. This dual-vessel approach reduces the required volume per vessel, enabling more compact equipment and shorter residence times while maintaining throughput. It also introduces redundancy into the system, allowing one vessel to remain operational while the other is cleaned, maintained, or cycled offline. This not only improves uptime but also ensures that gravity-driven flow can be maintained continuously, even during partial shutdowns.

In addition to gravity-based flow, the plant incorporates heat integration between process streams at different temperatures. For example, the exothermic leaching reactions involving sulfuric acid and cathode metals release significant heat. Rather than dissipating this energy, the hot effluent is routed through a heat exchanger to preheat incoming feed solutions or microbial media. This reduces the need for external heating utilities and improves the overall thermal efficiency of the process.

A third consideration for energy efficiency is the use of natural cooling towers to manage the temperature of non-critical aqueous streams. Instead of relying on electrically powered chillers, the plant uses passive cooling towers that leverage ambient airflow and evaporative cooling to reduce the temperature of raffinate and neutralized brine before discharge or reuse. This approach significantly lowers electricity consumption associated with thermal management and aligns with sustainable design principles.

Together, these strategies gravity-driven flow with redundant vessels, heat integration, and natural cooling form a cohesive energy conservation framework that supports both operational efficiency and environmental responsibility. Additional measures such as thermal insulation, vertical thermosiphons, and low-temperature microbial leaching further contribute to the plant's low-energy footprint.

7.5.2 Heat Exchange Fluids

Water was chosen as the heat-exchange medium for the reactor jacket because it provides the best overall balance of thermal performance, cost, safety, and environmental compatibility. For maximum Li recovery, the bioleaching process is to be maintained at 27 °C, which is within the liquid range of water. Specialty fluids such as Dowtherm or molten salts are formulated for high-temperature applications and would offer no advantage in this low-temperature regime. Water remains stable and non-volatile at these temperatures.

Water possesses one of the highest specific heat capacities of any common liquid (4.18 $kJ \cdot kg^{-1} \cdot K^{-1}$), which means it can absorb or release large amounts of energy per unit mass for small temperature changes. This enables precise temperature control of the reactor with minimal adjustments to the flow. Its high thermal conductivity ($\approx 0.6\ W \cdot m^{-1} \cdot K^{-1}$) further enhances heat transfer efficiency compared to most organic heat-transfer oils.

Water is easily accessible, inexpensive, and requires no special handling systems. Unlike specialty fluids, leak losses do not pose environmental hazards. Water is non-toxic, non-flammable, and chemically inert toward stainless steel and polymeric components at neutral to mildly acidic pH levels. This is important in a bioleaching plant, where safety and contamination control are critical.

Chapter 8. Project Management

8.1 Logistics

A robust logistics and sparing philosophy is essential to ensure the plant meets its operational target of 300 days per year. The logistics plan is designed to buffer the 24/7 continuous bioleaching process from disruptions in the feedstock supply chain, product offtake, or equipment failure. This is achieved through the strategic on-site storage of raw materials and products, as well as a defined equipment redundancy (spare) plan.

8.1.1 Raw Material and Product Storage

To ensure continuous operation, the plant will maintain a standard 14-day (2-week) storage capacity for all critical chemical raw materials and finished products. Feedstocks and intermediate buffers are designed to meet specific operational needs. This storage plan is segmented by material type. For feedstock, the receiving area will be designed to safely store a 14-day supply, equivalent to 700 tonnes of spent batteries (based on the 50 tonnes/day processing rate), in a designated, fire-rated, and access-controlled area. An intermediate Black Mass buffer silo will be located between the crushing circuit (CR-101) and the Black Mass Pump (P-102), holding a 48-hour (2-day) supply to decouple the mechanical and chemical circuits.

To support the chemical process, key reagents will also be stored on a 14-day basis. This includes a supply of solid sodium chloride ($NaCl$) for the Brine Solution Tank (V-100); dedicated, corrosion-resistant bulk storage tanks for Sulfuric Acid (H_2SO_4) (to be used by the Dosing Pump, P-203) and Sodium Hydroxide ($NaOH$) (for the SP-301 extractor); a silo for solid Sodium Carbonate (Na_2CO_3) (for the SP-302 circuit); and a bulk tank for the Cyanex 272 solvent, which holds the initial capital inventory plus a 14-day make-up supply for the SP-302 circuit.

Biologicals and utilities are managed differently. The Bacteria of A. Ferrooxidans will be stored as a concentrated culture in cooled totes, ready for inoculation of the Bioleach Vessels (V-101 A/B) via the Bioreactor Pumps (P-201). Water for heat exchange and general process use is sourced from the Water Reservoir (V-203) and circulated to the Heat Exchanger (F-201) to maintain optimal ambient temperature in the bioleach vessels.

Finally, the finished products warehouse will be designed to hold a 14-day inventory. Based on the total calculated output of 5,740.07 tonnes/year (approx. 19.1 tonnes/day), this equates to approximately 270 tonnes of packaged product (MHP, Li_2CO_3, etc.) collected from the filter presses (FP-301 to FP-305) and ready for shipment via the product conveyors (CO-301).

8.1.2 Equipment Redundancy and Replacement

To maximize plant availability (uptime) and ensure continuous operation, a standard N+1 redundancy philosophy is applied to all critical rotating equipment. This philosophy is evident in the design of the critical process pumps (e.g., P-101, P-102, P-201 to P-205) and the Air Compressors (C-101 A/B), which are all designed with an installed "standby" unit (N+1) to allow for immediate switch-over in the event of a failure or for planned maintenance.

Similarly, the process-critical reactor systems are designed with built-in redundancy to ensure continuous leachate flow. The Bioleach Vessels (V-101 A/B) and their associated Aerators (AE-201 A/B) are designed as two units of 100% capacity (N+N redundancy). This allows one entire leaching line to be taken offline for maintenance or cleaning while the other continues to operate at full plant capacity.

The logistics plan also accounts for the periodic replacement of components considered consumables. The primary consumables are the Filter Press Cloths for all filter presses (FP-301 through FP-305), which will be replaced based on pressure drop or performance decline. Secondary consumables include the Instrumentation Probes (pH and ORP sensors) in the abrasive, acidic environment of the bioleaching vessels, which will require frequent calibration and periodic replacement.

8.2 Plant Location

The selection of an optimal plant site is a critical decision that influences a project's long-term capital expenditure (CAPEX), operating expenditure (OPEX), logistical efficiency, and regulatory compliance. The siting of this bioleaching facility, designed to process 15,000 tonnes of spent batteries annually, requires a strategic balance of logistical, economic, and regulatory factors. This section outlines the methodology employed to evaluate candidate sites, including the qualitative and quantitative analysis of each location, as well as the process of final site selection.

8.2.1 Selection Methodology

To ensure a systematic, data-driven, and objective decision-making process, a comprehensive evaluation rubric was developed. This methodology involves four key steps:

1. Criteria Definition and Weighting: Key criteria were established across five major categories (Logistics, Economics, Labour, Regulatory, and Infrastructure). Each criterion was then assigned an "Importance Weight" on a scale of 1 (Low Importance) to 5 (Critical) to reflect its specific impact on the project's success.

2. Site Research: Detailed data was gathered for each potential location against the defined criteria.

3. Comparative Scoring: Each candidate site was rated on a scale of 1 (Poor Fit) to 5 (Excellent Fit) for each criterion.

4. Quantitative Analysis: The final decision is based on a weighted score, calculated by multiplying the "Importance Weight" by the "Location Score" for each criterion. The site with the highest aggregate weighted score is identified as the optimal choice.

8.2.2 Evaluation Criteria

The selection rubric was segmented into five primary categories, each containing specific, measurable criteria deemed essential for the plant's operational viability. The first category, Logistics & Supply Chain, assesses the efficiency of moving materials, including proximity to feedstock sources (such as spent battery hubs), access to multimodal transportation (highways, rail, ports), and proximity to end-markets (battery manufacturers, metal refineries). The second, Economics & Financials, evaluates the project's financial viability, considering key factors such as the cost of industrial-zoned land, construction costs, utility tariffs (especially for high-demand electricity), and the availability of government incentives, grants, or tax credits.

The third, Labour & Workforce, assesses the human resources environment, including the availability of skilled labour (such as chemical engineers, plant operators, and skilled trades) and the prevailing local wage rates for industrial workers. The fourth, Regulatory & Environmental, evaluates the legal and social operating landscape, including existing industrial zoning, the complexity and transparency of the environmental permitting process, the availability of specialized waste disposal infrastructure, and the level of community and political acceptance. Finally, Site Infrastructure evaluates the physical "at the boundary" readiness of the site, including access to high-capacity utilities (electricity, water, gas) and the geotechnical characteristics of the land (stability, contamination, topography).

8.2.3 Candidate Site Analysis

Three strategically distinct international locations were identified and evaluated: Bécancour, Québec (Canada); Skellefteå, Sweden; and Shenzhen, China.

8.2.3.1 Site A: Bécancour, Québec, Canada

Bécancour is identified as a central node in the emerging North American electric vehicle and battery-materials ecosystem, supported by a strong provincial government strategy. The site offers strong logistical advantages. It is situated on the St. Lawrence River, offering deep-water shipping access for bulk chemicals and products. Both CN and CP rail lines service the region, and it is connected via Autoroutes 55 and 40. Feedstock proximity is a key consideration; the site is located approximately 650 km from the Greater Toronto Area (GTA), a primary feedstock cluster, but is much closer to Montréal (150 km) and the U.S. border (200 km). This is within a single-day trucking radius of major supply hubs. The site is also in close proximity to a growing cluster of cathode, anode, and cell manufacturers in both Québec and Ontario, providing a strong local market for refined metals.

The primary economic driver for Bécancour is its access to Hydro-Québec's industrial power grid, which offers some of the lowest electricity rates in North America. This provides a significant and stable OPEX advantage. This is partially offset by moderately priced industrial land and high labour and material costs by global standards. The project would be eligible for federal and provincial programs, such as the Strategic Innovation Fund and Investissement Québec, which can offset capital costs through grants and tax credits.

A significant advantage is the availability of a skilled, bilingual workforce. The Montréal–Trois-Rivières–Québec City corridor hosts several universities (e.g., McGill, Polytechnique Montréal, UQTR) producing chemical and metallurgical engineers. The region already supports operators for existing aluminum smelters and chemical plants. While the labor pool is smaller than in a major urban core, industrial wages are moderate by North American standards.

From a regulatory and environmental standpoint, the site benefits from strong political and community support, as the Québec government actively promotes the "battery-materials corridor". This high community acceptance for green-tech projects is a major de-risking factor. The region has an established hazardous-waste management network stemming from its existing metallurgical industry, and its water resources, sourced from the St. Lawrence River, are abundant. Wastewater discharge is strictly regulated, which ensures environmental compliance. Finally, the Bécancour Industrial Park is pre-serviced with high-capacity utilities, including multiple high-voltage electrical feeds, industrial water networks, and natural gas. This "at the boundary" access eliminates the need for costly off-site utility extensions. The land is flat, geotechnically stable, and ideal for a large-footprint process plant.

8.2.3.2 Site B: Skellefteå, Sweden

Skellefteå represents a "green premium" strategy, offering deep integration into Europe's most advanced circular economy hub, anchored by the Northvolt Ett gigafactory. The site's primary logistical strength is its co-location with Northvolt. This provides a direct and consistent source of high-quality production scrap (a cleaner feedstock) and a built-in primary customer for the plant's refined metal outputs. This creates a near-perfect closed-loop system, eliminating significant transport costs. For other feedstock, the site relies on the mature European battery collection network, accessible via robust port, rail, and highway (E4) connections, though this may incur import duties.

Skellefteå's primary economic advantage is its access to abundant, 100% renewable electricity (hydro and wind) at rates among the lowest in Europe. This provides a major OPEX advantage and a powerful "green" marketing angle that can command premium product pricing. However, this is significantly offset by high Nordic land, construction, and labour costs. The project would be eligible for significant EU and Swedish grants for green technology and circular economy projects.

The site benefits from a highly skilled and educated Swedish workforce, supported by a strong engineering culture and excellent universities. The presence of Northvolt acts as a magnet for specialized talent in battery technology. This high-quality labour pool, however, comes at the cost of high industrial wages, which would increase operational expenditures. The regulatory environment is clear, stable, and supportive of the green industry. The precedent set by Northvolt means that the permitting pathways for this specific industry are well understood by local authorities. The site is subject to stringent EU standards for waste and wastewater, ensuring long-term compliance with these regulations. The infrastructure is excellent, with designated industrial areas pre-serviced with high-capacity electrical grids and other necessary utilities on geologically stable land.

8.2.3.3 Site C: Shenzhen, China

Shenzhen, situated in the Guangdong–Hong Kong–Macao Greater Bay Area, embodies a strategy centred on a global scale, market leadership, and unparalleled supply chain centrality. The logistical advantages of this site are unmatched. It is at the epicentre of global battery production, with China accounting for over 70% of global LIB production and Guangdong being a top province for both manufacturing and recycling. This provides an immediate and vast supply of feedstock from major producers like BYD and CATL. These same companies form a massive local customer base for refined metals. The site has access to world-class ports (Yantian, Shekou) and extensive highway and rail networks, facilitating both domestic and global logistics.

Economically, Shenzhen presents a trade-off. Industrial land is among the most expensive in China due to high demand and urban density. However, this high initial cost is strongly offset by competitive utility rates (e.g., ~US$0.085/kWh for industrial power) and significantly lower labour costs for both skilled and unskilled workers. Furthermore, the standard 25% corporate tax rate can be reduced to 15% in preferential zones, and various government subsidies are available for green tech and circular economy projects.

The region possesses one of the world's largest and most experienced concentrations of skilled industrial workers, technicians, and specialized engineers in electronics, battery chemistry, and chemical processing. Competition for talent is high, but the sheer size of the labour pool is a significant advantage for scaling operations. The regulatory environment is complex but well-defined within designated industrial parks and Special Economic Zones (SEZs). The Shenzhen Ecology and Environment Bureau has streamlined the permitting process for "collaborative approval."

Strict environmental controls for air and wastewater are mandatory, and a robust hazardous waste infrastructure is in place, with multiple licensed facilities capable of handling industrial sludges and heavy metal residues. Infrastructure within designated industrial parks is world-class, with sites pre-equipped with high-capacity connections to the advanced China Southern Power Grid, as well as industrial water and natural gas mains. The land is flat, stable, and engineered for heavy industry.

8.2.4 Comparative Analysis and Final Selection

Following the detailed analysis of each candidate site against the weighted criteria, a quantitative and qualitative assessment was performed to determine the final selection.

8.2.4.1 Quantitative Scoring Summary

A quantitative summary of the weighted scores for each site is presented below. The scores are calculated by multiplying the "Importance" (I) by the "Score" (S) for each of the 14 criteria, as detailed in the source rubrics.

Table 8.2: Comparative Decision Matrix for Plant Location

Site Location	Logistics Score (I*S)	Economics Score (I*S)	Labour Score (I*S)	Regulatory Score (I*S)	Infrastructure Score (I*S)	Total Score
Bécancour, QC	45.5	37.0	28.5	42.5	29.0	182.5
Skellefteå, SE	40.0	42.0	16.5	39.5	30.0	168.0
Shenzhen, CN	60.0	42.5	35.0	33.5	30.0	201.0

8.2.4.2 Qualitative Rationale

The quantitative analysis indicates a clear preference for Shenzhen, China, which achieved the highest weighted score (201.0) based on the assigned importance weights and research data.

Shenzhen's winning performance is attributed to its unparalleled logistical advantages (a perfect weighted score of 60.0) and superior labour metrics (a perfect score of 35.0). Its position as the global hub for battery production provides an unmatched concentration of both feedstock supply and end-market customers. Furthermore, highly competitive labour and utility costs provide a strong foundation for long-term operational profitability that offsets the high initial land cost.

Bécancour, QC, performed exceptionally well as a strong second choice (182.5). Its primary strengths are its access to extremely low-cost renewable power and a highly supportive regulatory and community environment. It represents a stable, sustainable, and strategically sound alternative for the North American market.

Skellefteå (168.0), while offering outstanding green credentials and infrastructure, was significantly penalized in the quantitative analysis due to its high labour costs and smaller, competitive labour pool (weighted score of 16.5). This made it the least economically viable option of the three, despite its strong strategic fit within the Northvolt ecosystem.

8.2.4.3 Final Site Selection

Based on a comprehensive quantitative and qualitative analysis, Shenzhen, China, has been selected as the designated location for the bioleaching battery recycling plant. The decision is primarily driven by the site's unmatched centrality in the supply chain, which provides direct access to the world's largest and most concentrated pool of spent batteries and production scrap.

This logistical advantage is complemented by proximate end-markets, as the site is co-located with the world's largest battery manufacturers, providing a ready and substantial customer base. These factors, combined with a superior cost-benefit profile driven by favourable labour and utility costs and strong economic incentives, are projected to provide the best long-term return on investment for a 15,000-tonne-per-year facility operating on a global scale.

8.2.5 Aerial View

The proposed Shenzhen lithium-ion battery recycling plant has been designed in accordance with the principles of inherently safe design, functional zoning, and efficient logistics. Raw material receiving, central processing, monitoring/control, and product shipping are each located in clearly delineated areas to minimize cross-traffic and reduce the risk of contamination. A ring road and branch access routes ensure smooth movement of vehicles and provide emergency access to all zones. This arrangement reflects best practice guidance for chemical plant siting and layout, which emphasizes separation of hazardous operations, streamlined material flow, and provision of safety buffers.

Figure 8.1: Aerial Visualization of Plant Design Concept

Figure 8.1 is a visualization of the aerial layout of the proposed lithium-ion battery recycling plant in Shenzhen, China. The Raw Materials Receiving is located at the front of the site for direct truck access. Includes unloading bays, a weighbridge, and segregated storage for different battery chemistries. This aligns with best practice to minimize transport distances and keep hazardous materials away from populated areas (HSE, n.d.). Processing Zone placed at the center of the facility to act as the operational core. Houses, shredding, bio-leaching reactors, purification tanks, and precipitation units. Surrounded by a loop road for maintenance and emergency access. Centralizing process units reduces piping runs and improves operability (Toghraei, 2015). Monitoring & Control is located adjacent but offset from the processing area. It contains the control room, laboratories, and administrative offices. While providing oversight and maintaining a safety buffer from hazardous operations (Chemical Times, 2024). Products Shipping is positioned on the opposite side of the site from raw materials receiving. Outbound logistics area with loading docks and truck bays. This separation prevents cross-traffic and reduces congestion, a principle emphasized in modern plant layout design (Piping-World, n.d.).

Pathways and Roads: The main ring road encircles the processing zone, ensuring that emergency vehicles can reach all areas. Branch roads connect receiving and shipping zones, keeping inbound and outbound traffic separate. Pedestrian pathways link monitoring/control to processing and shipping, ensuring safe operator movement. Recommended safety distances and access routes are consistent with CCPS siting guidelines (Aisyah, 2021). Pedestrian pathways link the monitoring/control building to processing and shipping areas. Green buffer zones surround the facility to provide environmental shielding and emergency access corridors.

The plant layout features green buffer zones around its perimeter, which reduce noise, capture dust, and serve as a visual barrier between industrial operations and the surrounding urban environment. Stormwater and wastewater are routed to a dedicated treatment unit, ensuring that acidic or alkaline effluents are neutralized before discharge, in line with Chinese environmental standards for industrial parks. Air emissions from shredding and leaching are controlled through local exhaust ventilation, HEPA filtration, and acid mist scrubbers, minimizing the release of particulates and vapours. Noise control measures include acoustic insulation in shredding halls and the installation of silencers on blowers. Together, these features demonstrate that the plant design integrates environmental protection with process safety, supporting both regulatory compliance and community acceptance.

Shenzhen provides a highly advantageous location for establishing a processing plant due to its proximity to Hong Kong and its integration within the Greater Bay Area's transport and logistics network. The Qianhai district in particular has been positioned as a hub for cross-border cooperation, with planned expressways and intercity rail links directly connecting it to Hong Kong International Airport, one of the busiest air cargo hubs in the world (Qianhai Authority, 2023). This ensures rapid access to global markets for high-value products while also supporting efficient inbound logistics for raw materials.

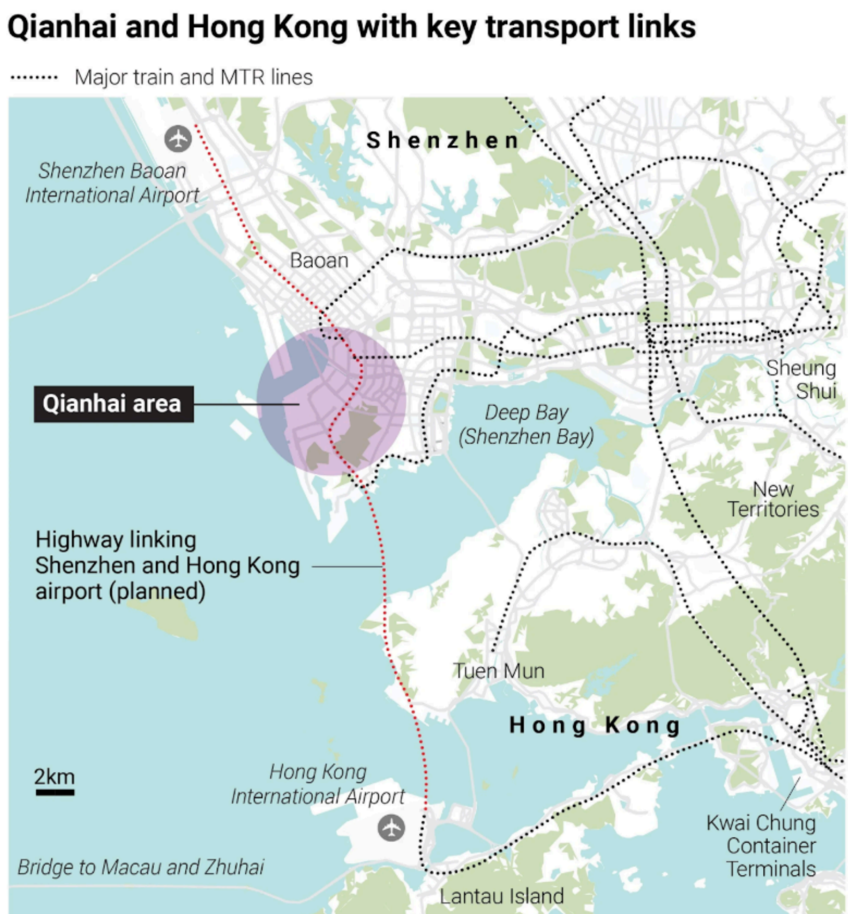

Figure 8.2: Transport Connectivity between Shenzhen and Hong Kong in the Greater Bay Area (adapted from Cheung, T. (2021))

As seen in Figure 8.2, in addition to Hong Kong's airport, Shenzhen itself is home to Baoan International Airport, which has rapidly expanded its cargo capacity to exceed 900,000 tons annually, driven by e-commerce and high-tech exports (Unitex Logistics, 2024; Shenzhen Daily, 2024). The integration of Qianhai's bonded logistics zone with Shenzhen Airport further streamlines customs clearance and international freight handling (Shenzhen Airport, 2022). This dual-airport advantage provides redundancy and flexibility for international shipments.

Shenzhen's coastal location also places it adjacent to some of the world's busiest container ports, offering direct maritime access to global shipping routes. Combined with the Hong Kong–Zhuhai–Macao Bridge and the Guangzhou–Shenzhen–Hong Kong Express Rail

Link, the city is integrated into a multimodal transport system that reduces travel times and enhances regional integration (Hong Kong SAR Government, 2024). This infrastructure not only facilitates the movement of goods but also supports workforce mobility and industrial clustering across the Greater Bay Area.

Finally, coordinated infrastructure investment between Hong Kong and Shenzhen has been explicitly designed to enhance industrial integration and cross-border logistics efficiency (EY, 2022). This positions Shenzhen as a strategic gateway for both domestic Chinese supply chains and international trade. In summary, Shenzhen's dual access to world-class airports, proximity to Hong Kong, adjacency to major seaports, and integration into the Greater Bay Area's transport corridors make it an ideal location for a plant requiring both international connectivity and regional industrial support.

Chapter 9. Conclusions

During the plant design project, a comprehensive review of existing lithium-ion battery recycling technologies was conducted, followed by a rigorous selection process for the optimal recovery method. Alternative approaches included pyrometallurgy, hydrometallurgy, and direct regeneration. Biohydrometallurgy was chosen for its superior environmental performance and technical viability. All process equipment was specified based on detailed flowsheet development and operational constraints identified for the recycling of 15,000 tonnes of spent batteries annually. Thorough safety analyses were performed to ensure compliance with regulatory and operational standards.

Subsequent mass and energy balances were completed for the entire plant, resulting in the determination that the facility produces lithium carbonate at a rate of 270.8 kg/h, maintained across continuous twenty-four-hour operation each month. This output reflects the integration of reactor controls, separation units, and downstream purification steps essential to high-purity product recovery. The project's conclusion also included logistical planning and plant location selection, with final recommendations based on efficiency, sustainability, and market demand. Ultimately, these achievements demonstrate the successful design and analysis of a next-generation battery recycling facility capable of achieving both economic and environmental objectives.

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Appendix A. Material Safety Data Sheets

Table A.1: Material Safety Data Overview for Plant Chemicals

Chemical / Material	Key Hazards	PPE / Controls	Storage & Incompatibilities	First Aid / Emergency Response
Sulfuric Acid (H_2SO_4)	Corrosive; exothermic with water; SO_x release	Acid-resistant gloves, face shield, apron; fume hood	Segregate from bases, cyanides, carbides, organics; corrosion-resistant containment	Flush skin/eyes ≥15 min; remove clothing; seek medical help; neutralize spills with sodium carbonate
Ferric / Ferrous Sulfate ($Fe_2(SO_4)_3$ / $FeSO_4$)	Irritant; dust causes eye/respiratory irritation; environmental hazard	Safety glasses, gloves; dust control	Keep dry; separate from strong bases/oxidizers	Rinse eyes/skin with water; fresh air if inhaled; sweep spills carefully
Hydrogen Peroxide (H_2O_2, 3–35%)	Strong oxidizer; violent decomposition with contaminants; burns	Goggles, face shield, gloves; vented storage	Segregate from organics, metals, reducing agents; temperature-controlled	Flush with water; fresh air if inhaled; dilute spills with water
Biogenic / Organic Acids (acetic, gluconic, microbial acids)	Corrosive/irritant; lower hazard than mineral acids	Safety glasses, gloves; splash protection	Avoid mixing with strong oxidizers/bases	Rinse skin/eyes; neutralize spills with sodium bicarbonate
Sodium Hydroxide (NaOH)	Strong caustic; exothermic dissolution; severe burns	Alkali-resistant gloves, goggles/face shield	Keep dry; segregate from acids, ammonium salts, aluminum	Flush with water ≥15 min; remove clothing; seek medical help; neutralize spills with dilute acid

Sodium Carbonate (Na_2CO_3)	Irritant; CO_2 release on neutralization	Safety glasses, gloves; dust control	Separate from acids until controlled addition	Rinse eyes/skin; sweep spills carefully
NaOH + Na_2CO_3 Systems	Heat generation, aerosol formation, scaling	Closed addition, temperature monitoring	Maintain clean lines; avoid uncontrolled mixing	Flush exposures with water; neutralize spills carefully
Solvent Extraction Reagents (DEHPA, Cyanex 272 + kerosene diluent)	Flammable; irritant; environmental hazard; vapors from diluent	Solvent-resistant gloves, goggles; explosion-proof ventilation	Segregate from oxidizers; ground/bond containers	Fresh air if inhaled; wash skin; absorb spills with inert material; eliminate ignition sources
Sodium Chloride Brine (NaCl solution)	Low hazard; electrical safety with live cells	Eye protection, gloves; insulated tools	Standard aqueous storage; avoid energized equipment	Rinse skin/eyes; mop spills; ensure electrical isolation
Nitrogen / Argon (inert gases)	Asphyxiation in confined spaces; pressure hazards	O_2 monitoring; cylinder handling training	Secure cylinders; keep away from heat	Move to fresh air; administer oxygen if trained; call emergency services
Microbial Media (e.g., 6K medium)	Low hazard; BSL-1 organisms; aerosols may irritate	Lab coat, gloves, eye protection	Store sterile; dispose via autoclave	Wash skin; flush eyes; disinfect spills with bleach

Metal Product Salts (Li_2CO_3, $Co(OH)_2$, $Ni(OH)_2$, $MnCO_3$, $FePO_4 \cdot 2H_2O$)	Dust inhalation; irritants; Ni/Co compounds may be sensitizers/carcinogens	Dust masks/respirators; local exhaust; closed handling	Dry, sealed containers; separate from acids/bases	Rinse skin/eyes; fresh air if inhaled; HEPA vacuum spills

1. Sulfuric Acid (H_2SO_4)

Identification: Strong mineral acid used in leaching. Hazards: Corrosive to skin/eyes; exothermic with water; SO_x release with incompatibles. PPE: Acid-resistant gloves, face shield, apron; fume hood. Storage: Segregate from bases, cyanides, carbides, organics; corrosion-resistant containment. First Aid: Flush skin/eyes ≥15 min; remove clothing; seek medical help; neutralize spills with sodium carbonate.

2. Ferric Sulfate ($Fe_2(SO_4)_3$) / Ferrous Sulfate ($FeSO_4$)

Identification: Iron salts used as oxidants/mediators. Hazards: Irritant; dust causes eye/respiratory irritation; environmental hazard in large releases. PPE: Safety glasses, gloves; dust control. Storage: Keep dry; separate from strong bases/oxidizers. First Aid: Rinse eyes/skin with water; fresh air if inhaled; sweep spills carefully.

3. Hydrogen Peroxide (H_2O_2, 3–35%)

Identification: Oxidant for Mn precipitation. Hazards: Strong oxidizer; violent decomposition with contaminants; burns. PPE: Goggles, face shield, gloves; vented storage. Storage: Segregate from organics, metals, reducing agents; temperature-controlled. First Aid: Flush with water; fresh air if inhaled; dilute spills with water.

4. Biogenic/Organic Acids (Acetic, Gluconic, microbial acids)

Identification: Produced by microbes or added as leaching aids. Hazards: Corrosive/irritant; lower hazard than mineral acids. PPE: Safety glasses, gloves; splash protection. Storage: Avoid mixing with strong oxidizers/bases. First Aid: Rinse skin/eyes; neutralize spills with sodium bicarbonate.

5. Sodium Hydroxide (NaOH)

Identification: Strong base for precipitation. Hazards: Strong caustic; exothermic dissolution; severe burns. PPE: Alkali-resistant gloves, goggles/face shield. Storage: Keep dry; segregate from acids, ammonium salts, and aluminum. First Aid: Flush with water ≥15 min; remove clothing; seek medical help; neutralize spills with dilute acid.

6. Sodium Carbonate (Na_2CO_3)

Identification: Base for lithium carbonate precipitation. Hazards: Irritant; CO_2 release on neutralization. PPE: Safety glasses, gloves; dust control. Storage: Separate from acids until controlled addition. First Aid: Rinse eyes/skin; sweep spills carefully.

7. Solvent Extraction Reagents (DEHPA, Cyanex 272 with kerosene diluent)

Identification: Organic extractants for cobalt separation. Hazards: Flammable; irritant; environmental hazard; vapors from diluent. PPE: Solvent-resistant gloves, goggles; explosion-proof ventilation. Storage: Segregate from oxidizers; ground/bond containers. First Aid: Fresh air if inhaled; wash skin; absorb spills with inert material; eliminate ignition sources.

8. Sodium Chloride Brine (NaCl solution)

Identification: Used in discharge and neutralization. Hazards: Low hazard; electrical safety with live cells. PPE: Eye protection, gloves; insulated tools. Storage: Standard aqueous storage; avoid energized equipment. First Aid: Rinse skin/eyes; mop spills; ensure electrical isolation.

9. Nitrogen / Argon (Inert gases)

Identification: Used as inert atmosphere. Hazards: Asphyxiation in confined spaces; pressure hazards. PPE: O_2 monitoring; cylinder handling training. Storage: Secure cylinders; keep away from heat. First Aid: Move to fresh air; administer oxygen if trained; call emergency services.

10. Microbial Media (e.g., 6K medium for *Acidithiobacillus*)

Identification: Nutrient medium for bioleaching microbes. Hazards: Low hazard; BSL-1 organisms; aerosols may irritate. PPE: Lab coat, gloves, eye protection. Storage: Store sterile; dispose via autoclave. First Aid: Wash skin; flush eyes; disinfect spills with bleach.

11. Metal Product Salts (Li_2CO_3, $Co(OH)_2$, $Ni(OH)_2$, $MnCO_3$, $FePO_4 \cdot 2H_2O$)

Identification: Final recovered products. Hazards: Dust inhalation; irritants; Ni/Co compounds may be sensitizers/carcinogens. PPE: Dust masks/respirators; local exhaust; closed handling. Storage: Dry, sealed containers; separate from acids/bases. First Aid: Rinse skin/eyes; fresh air if inhaled; HEPA vacuum spills.

Appendix B. Sample Calculations

<u>Mass Balance Sample Calculations:</u>

<u>Conversion of mass flow for the battery input from Project 12 Statement:</u>

$$m_{in} = \frac{15000ton}{year} * \frac{1000kg}{ton} * \frac{year}{300day} * \frac{day}{24hr} *= 2083.33kg/h\ of\ batteries\ processed\ every\ day$$

<u>Calculating the total mass flow of brine solution,</u>

Given a 3:1 Brine to Battery mass ratio,

$$m_{brine,\,total} = 2083.33\ kg/hr \times 3 = 6250\ kg/hr\ of\ Brine\ solution\ in\ circulation$$

Taking Brine Density as 1050 kg/m^3 and a 15% purge per day, the rate of Brine solution being used up per hour is,

$$m_{brine,\,used} = 6250\ kg/hr \times 15\% \times \frac{1}{24} = 39.0625\ kg/h\ of\ Brine\ solution$$

<u>Control Volume of Crusher Mass Balance:</u>

For the crusher recovery of metal, based on Garcia et al. (2025), the percent recovery is as follows, obtained in an unsorted black mass of metal composition.

<u>Control Volume of CSTR Mass Balance:</u>

Given that the Lithium is 2.80% of the total battery weight,

$$m_{Li} = 2.80\% \times Battery\ Mass = 2.80\% \times 2083.33\ kg/hr = 58.33\ kg/hr.$$

<u>Determining Lithium recovery rate,</u>

Given that the bioleaching process has a 90% Lithium recovery rate, Garcia et al. (2025)

$$m_{Li,\,rec} = 90\% \times m_{Li} = 90\% \times 58.33\ kg/hr. = 52.50\ kg/hr.$$

<u>Determining the required mass flow of the Culture Medium needed,</u>

Using a 1% weight-to-volume ratio of 1kg to 100 L from the CSTR operation,

$$sum\ of\ all\ material\ present\ in\ CSTR * 100 = 935.4167 kg/hr * 100 = 93541.67 kg/hr$$

<u>Mass Balance and Composition of the Culture Medium (CM) present to promote bio-reaction in the CSTR:</u>

$$m_{CMcomp} = Composition\ density\ of\ material * mass\ flow\ CM$$

$$m_{(NH_4)_2SO_4} = 3 * 1000 kg/L * 93541.64 kg/h = 280.62 kg/h\ of\ (NH_4)_2SO_4$$

<u>Determination of waste/unreacted metals:</u>

$$Mass\ flow\ of\ Input\ metals - Mass\ flow\ of\ output\ metals = Waste\ of\ unreacted\ metals$$

$$Waste\ of\ unreacted\ metals = 40.06 kg/h$$

All other required chemicals are calculated using the same method for Separator 1 to obtain the following massflow needed:

$$Mass\ of\ needed\ Na^{+} = 6.75 kg\ or\ more\ in\ solution$$

$$Mass\ of\ needed\ Fe^{0} = 193.56 kg\ of\ steel\ wool$$

<u>Determining Sodium Hydroxide based on the desired PH required to separate metals after the bioleaching process:</u>

$$Sum\ of\ all\ mass\ flow\ in\ Separator\ 1 = 94477.1 kg\ of\ black\ mass\ solution\ per\ batch$$

$$Desired\ PH\ of\ 2 = (10^{-Current\ PH} - 10^{-Desired\ PH}) * 94477.1 kg$$

$$Desired\ PH\ of\ 2 = (10^{-1.5} - 10^{-2}) * 94477.1 kg = 2042.86 mol\ of\ NaOH$$

$$Desired\ PH\ of\ 2 = 2042.86 mol\ of\ NaOH * 40 g/mol * 1000 kg/g = 81.71 kg\ of\ NaOH$$

81.71kg of NaOH is needed to raise the pH to 2, which precipitates out 100% of the Iron present in the separation vessel. Raising the pH to 3 precipitates Manganese out at 85% recovery, requiring 34.01kg of H2O2. Raising the pH to 4.5 precipitates Aluminum out at 95% recovery, requiring 0.412kg of NaOH.

<u>From Separator 1, precipitates that can be filter-pressed out are Al, Fe, Mn, Cu; as such, we can determine the mass flow rate recovered from Filter Press 1:</u>

$$m_{wet\ metal} = m_{metal} * recovery\%\ of\ separator\ 1$$

$$m_{wet\ Al} = 152kg/h * 0.95 = 144.4kg/h\ of\ wet\ Al\ obtained\ from\ separator\ 1$$

<u>To obtain dry metal from the Filter Press, there is a slight loss in the obtained material:</u>

$m_{dry\ metal} = m_{wet\ metal} * recovery\%\ of\ filter\ press\ 1$ (Suez Water handbook)

$$m_{wet\ Al} = 144.4kg/h * 0.9999 = 142.95kg/h\ of\ dry\ Al\ obtained\ from\ filter\ press\ 1$$

<u>Mass Balance on Separator 2 used precipitate out Li, Co, Ni, Cu:</u>

$$Sum\ of\ all\ mass\ flow\ output\ in\ Filter\ Press\ 1\ =\ kg\ of\ black\ mass\ solution\ input\ Separator\ 2$$

Determining amount of required Cyanex-272 to leach out the most Cobalt

$$Required\ Cyanex\ moles\ =\ 2 * \frac{Mass\ flow\ of\ Cobalt}{Molar\ Mass\ of\ Cobalt} = 2 * Cobalt\ input\ of\ separator\ 2$$

$$Cyanex\ required\ in\ kg\ =\ required\ Cyanex\ moles\ *\ Molar\ mass\ of\ Cyanex$$

$$m_{Cyanex} = 1.5612kmol/h * 290.4kg/kmol = 453.36kg/h$$

Determining amount of unreacted Cyanex-272

$$Fraction\ of\ reacted\ Co = \frac{(Distribution\ Ratio)*(Phase\ Ratio)}{1+(Distribution\ Ratio)*(Phase\ Ratio)} = \frac{10*1}{1+10*1} = 0.91 = 91\%$$

$$Moles\ Unreacted\ Co\ =\ (100\%\ -\ 91\%)\ *\ (moles\ inputted\ Co)\ =\ (9\%)(0.781)\ =\ 0.7096 kmol/h$$

$$Moles\ Cyanex\ Unreacted\ =\ 2\ *\ Unreacted\ moles\ Co\ =\ 2\ *\ 0.7096 kmol/h\ =\ 1.419 kmol/h$$

All other required chemicals are calculated using the same method to obtain the following massflow needed:

$$Mass\ of\ needed\ Na^{+}\ =\ 170.84 kg\ of\ Na_2CO_3$$

Same process of recovery using Filter Press 2 is achieved with differing recovery rates:

Energy Balance Sample Calculations:

Average Temperature per month

$$Average\ month\ temp.\ =\ \frac{temp.\,high + temp.\,low}{2} = \frac{19+11}{2} = 15°C$$

Bioleaching Vessel Water Temperature Needed for Operation at 27°C

$$Q_{jacket\ needed} = feed\ energy\ +\ air\ energy\ +\ evap(cooling)\ +\ agitator\ energy$$

$$+\ rxn\ energy\ +\ wall\ loss/gain$$

$$Feed\ Energy\ = \dot{m}_f c_{p,f}(T_{feed} - T_r)\ =\ 0.133\frac{kg}{s} * 0.7\frac{kg}{kJ*K} * (15°C\ -\ 27°C) =-\ 1.12kW$$

$$air\ energy\ = \dot{m}_{air} c_{p,air}(T_{air} - T_r)\ =\ 0.1\frac{kg}{s} * 0.718\frac{kg}{kJ*K} * (15°C\ -\ 27°C) =-\ 0.86kW$$

$$evap(cooling)\ = \dot{m}_{evap}\Delta H_{vap}(T_r)\ =\ (10\frac{kg}{s} * (Y_{out} - Y_{in})\frac{1}{3600}) * 0.718\frac{kg}{kJ*K} * 15°C\ =\ 0.015kW$$

$$rxn\ energy\ =\ V\ * \sum_i r_i * (\Delta H_i)\ =\ 50m^3 * 0.0004861\frac{mol}{m^3 s} * 98\frac{kJ}{mol}\ =\ 2.38189kW$$

$$wall\ loss/gain\ =\ UA(T_a - T_r)\ =\ 3.7\frac{kW}{K}(15°C\ -\ 27°C) =-\ 44.4kW$$

$$Q_{jacket\ needed} =- 1.12kW +- 0.86kW + 0.015kW + 2.38189kW +- 44.4kW = 23.98kW$$

$$T_{water\ needed} = T_{needed} + \frac{Q}{UA} + \frac{Q}{2*\dot{m}*c_p} = 27°C + \frac{23.98kW}{3.7\frac{kW}{K}} + \frac{23.98kW}{2*1\frac{kg}{s}*4.186\frac{kJ}{kg*K}} = 36.35°C$$

<u>Separator 2 Jacket Energy Needed for Operation at 60°C:</u>

$$Q_{jacket\ needed} = Feed\ Energy + wall\ loss/gain = \dot{m} * c_p * (T_{feed} - T) + UA(T_{air} - T)$$

$$Q_{jacket\ needed} = [26.23kg/s * 4.1\frac{kJ}{kg*K} * (27 - 60)] * \frac{1}{1000} + 3.7\frac{kW}{K}(15 - 60) = 170kW$$

<u>Compressor Energy:</u>

Based on the experiment conducted in Garcia et al. (2025), the compressor was operating at (50Nhl/h)/2L

$$Scaled\ up\ aeration = V_{new} * \frac{50\frac{Nhl}{h}}{2L} = 50m^3 * 25\frac{\frac{m^3 air}{h}}{m^3} = 1250\frac{m^3 air}{h}$$

$$Power\ needed = \frac{scaled\ up\ aeration*\Delta P}{\eta} = 2 * \frac{1250\frac{m^3 air}{h}*\frac{1h}{3600s}*2*10^5 Pa}{0.7} = 200kW$$

<u>Pump Energy:</u>

$$Energy = density * g * volumetric\ flow * dynamic\ head = \rho gQH$$

$$Energy = \dot{m}gH = (0.011kg/s) * (9.81\frac{m}{s^2}) * (1m) = 0.106kW$$

<u>Conveyor Energy:</u>

$$P_{lift} = \dot{m}gH = (0.579\frac{kg}{s}) * (9.81\frac{m}{s^2}) * (5m) = 28.39W$$

$$P_{roll,mat} = \dot{m}gf_r L = (0.579\frac{kg}{s}) * (9.81\frac{m}{s^2}) * (0.025) * (5m) = 0.7096W$$

$$P_{roll,belt} = m_b^{'} L g f_r v = (17.5\frac{kg}{m}) * (5m) * (9.81\frac{m}{s^2}) * (0.025) * 0.1\frac{m}{s} = 2.146W$$

$$P_{shaft} = P_{lift} + P_{roll,mat} + P_{roll,belt} = 28.39W + 0.7096W + 2.146W = 31.24W$$

$$P_{motor} = \frac{P_{shaft}}{\eta} * \frac{1kW}{1000W} = \frac{31.24W}{0.9} * \frac{1kW}{1000W} = 0.03471kW$$

<u>Heat Exchanger Energy:</u>

$$Q = \dot{m}c_p(T_{out} - T_{in}) = (1\frac{kg}{s})(4.186\frac{kJ}{kg*K})(36.25°C - 15°C) = 89.35kW$$